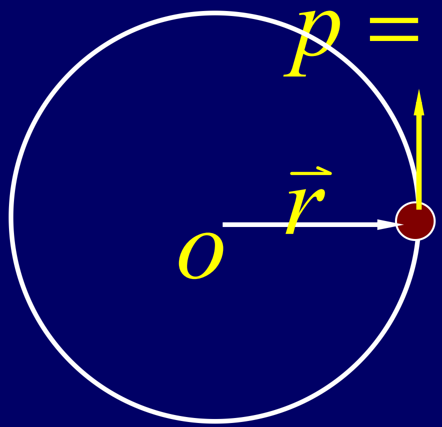
Chapter 10
Angular Momentum

Angular momentum of a particle

$$\vec{p} = m\vec{v}$$



The angular momentum with respect to O

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L = rp\sin\theta$$

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}$$

$$= \vec{v} \times m\vec{v} + \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt}$$

$$= \vec{r} \times \vec{F}$$

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$

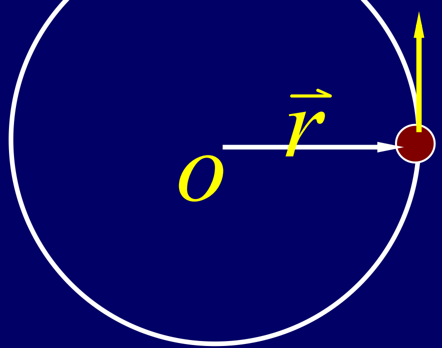
Angular momentum

$$\frac{d\vec{p}}{dt} = \vec{F}$$

linear momentum

Angular momentum of a particle

$$\vec{p} = m\vec{v}$$



The angular momentum with respect to O

$$\begin{cases} \dfrac{dp_x}{dt} = F_x \\ \dfrac{dp_y}{dt} = F_y \\ \dfrac{dp_z}{dt} = F_z \end{cases} \qquad \begin{cases} \tau_x = \dfrac{dL_x}{dt} \\ \tau_y = \dfrac{dL_y}{dt} \\ \tau_z = \dfrac{dL_z}{dt} \end{cases}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

$$L = rp\sin\theta$$

$$\frac{d\vec{L}}{dt} = \vec{\tau}$$ **Angular momentum**

$$\frac{d\vec{p}}{dt} = \vec{F}$$ **linear momentum**

Both $\vec{\tau}$ and $\mathbf{L}$ must be with respect to the same origin.

Impulse *and* ***momentum***

$$\int_0^t \Sigma \vec{F} dt = \Delta \vec{p} = \vec{p}_f - \vec{p}_i$$

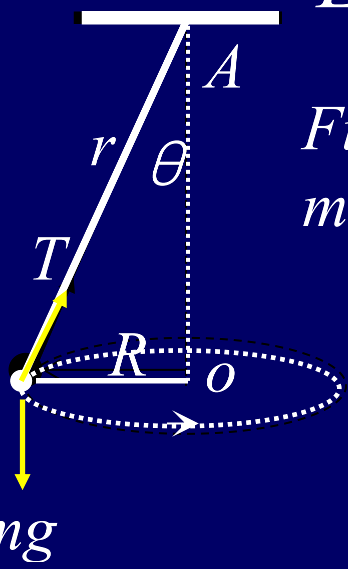
Angular impulse and **angular momentum**

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt} \quad \sum \vec{\tau} dt = d\vec{L} \quad \int_0^t \sum \vec{\tau} dt = \Delta \vec{L} = \vec{L}_f - \vec{L}_i$$

$$\int_0^t \sum \vec{\tau} dt = 0, \quad \vec{L}_f - \vec{L}_i = 0 \quad \vec{L}_f = \vec{L}_i$$

*Both $\vec{\tau}$ and **L** must be with respect to the same origin.*

Example:

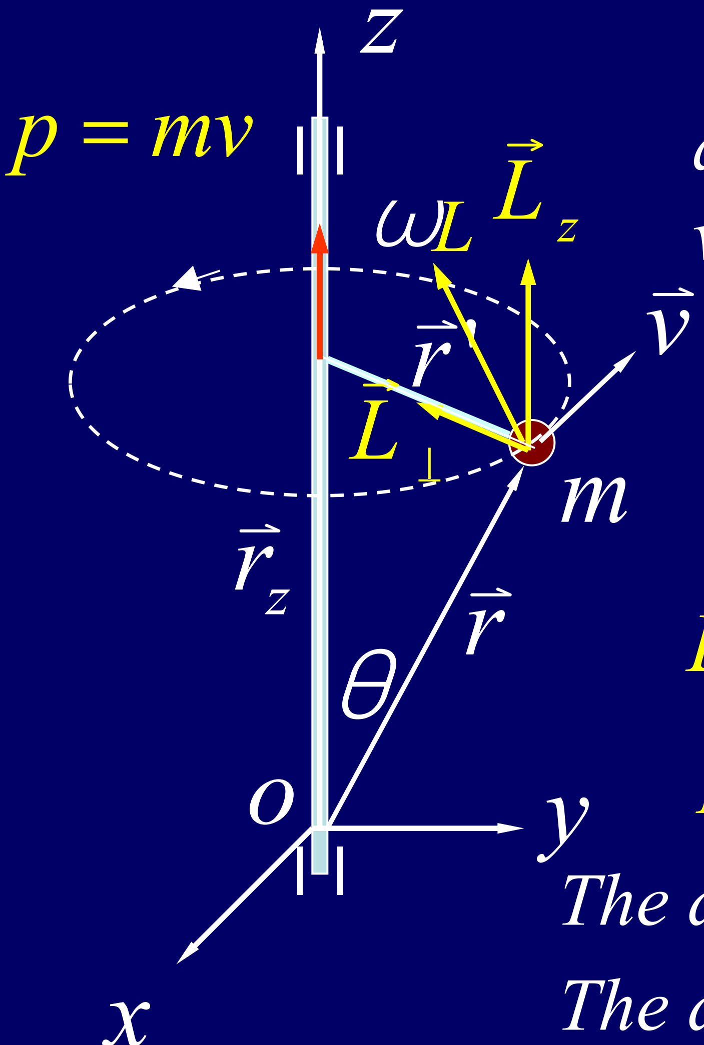


Find the torque of T and mg, total torque, angular momentum with respect to points A and O

τ_T	τ_{mg}	τ_{T+mg}	L
$\vec{R}\times\vec{T}$	$\vec{R}\times m\vec{g}$	$\vec{R}\times(\vec{T}+m\vec{g})$	$\vec{R}\times m\vec{v}$
$RT\cos\theta$	Rmg	0	Rmv
$\otimes$	$\odot$		$\uparrow$
$\vec{r}\times\vec{T}$	$\vec{r}\times m\vec{g}$	$\vec{r}\times(\vec{T}+m\vec{g})$	$\vec{r}\times m\vec{v}$
0	$rmg\sin\theta$	$rmg\sin\theta$	rmv
	Rmg	Rmg	
	$\odot$	$\odot$	$\nwarrow$

The angular momentum for the point A is not a constant vector because its direction changes, but for the point O it is a constant vector.

Angular momentum and angular velocity



A particle of mass m rotates about the z axis with angular velocity ω.

The tangential speed is v=ωr´.

$$\vec{L} = \vec{r} \times \vec{p} = (\vec{r}_z + \vec{r}\,') \times \vec{p}$$

$$= \vec{r}_z \times \vec{p} + \vec{r}\,' \times \vec{p} = \vec{L}_\perp + \vec{L}_z$$

$$L_z = r'mv = r'm\omega r' = mr'^2\omega = I_z\omega$$

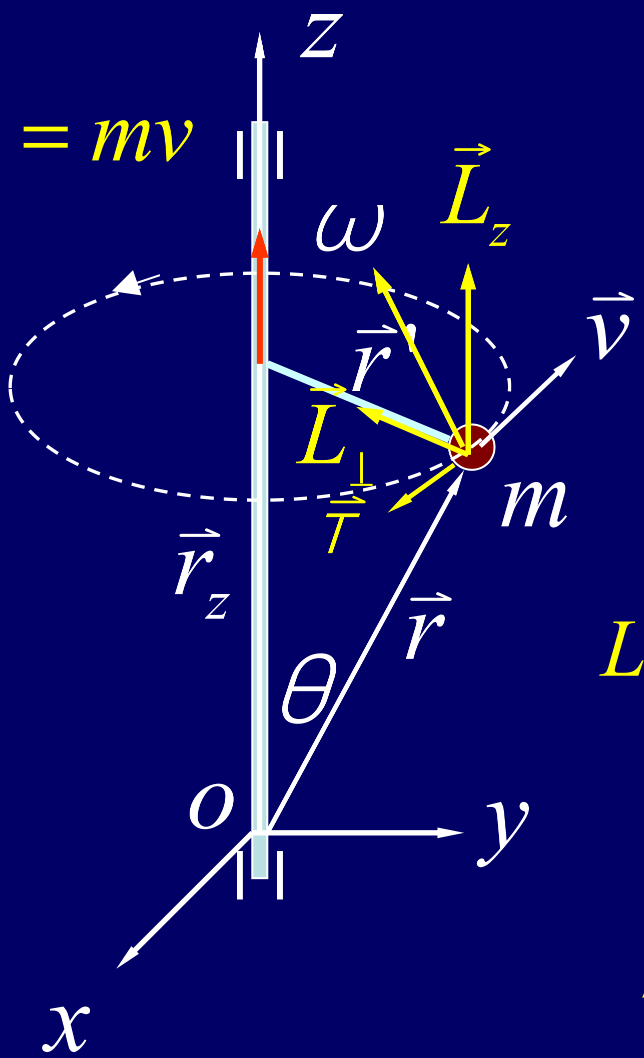
$$\vec{p} = m\vec{v} \quad \vec{L} \neq I\vec{\omega} \qquad L_z = I_z\omega$$

The direction of ***L*** *is not the same as* ***ω****.*

The direction of ***L****$_z$* *is the same as* ***ω****.*

Angular momentum and angular velocity

$p = mv$



A particle of mass m rotates about the z axis with angular velocity $\boldsymbol{\omega}$*. The tangential speed is v=ωr´.*

$$\vec{L} = \vec{r} \times \vec{p} = (\vec{r}_z + \vec{r}\,') \times \vec{p}$$

$$= \vec{r}_z \times \vec{p} + \vec{r}\,' \times \vec{p} = \vec{L}_\perp + \vec{L}_z$$

$$L_\perp = r_z m v = r_z m \omega r' = (mr\cos\theta)(\omega r\sin\theta)$$

$$= mr^2\omega\cos\theta\sin\theta$$

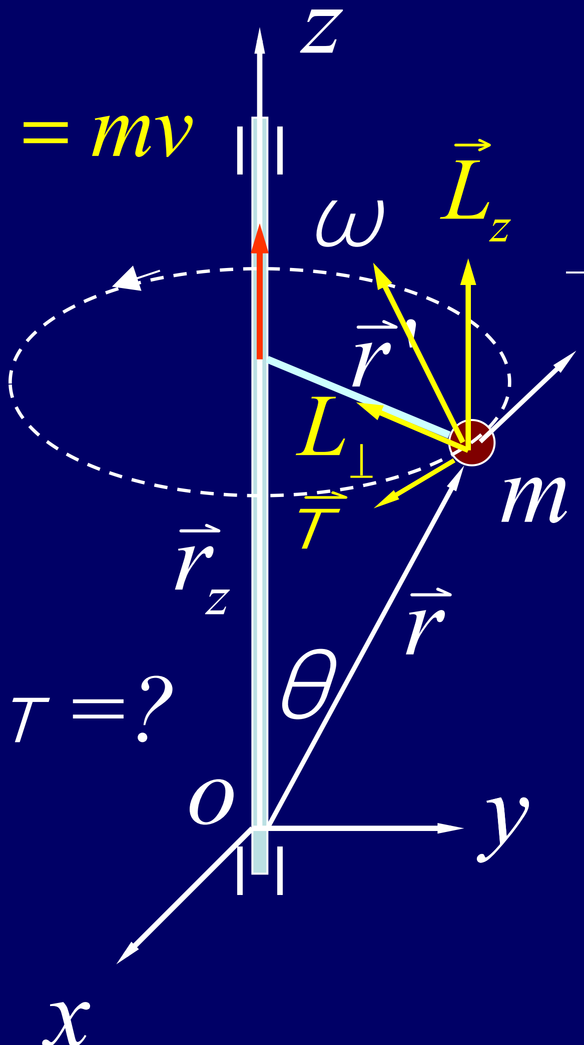
$$\vec{\tau} = \vec{r} \times \vec{F} \qquad F = m\omega^2 r'$$

$$\tau = rF\sin(\pi/2 + \theta) = rm\omega^2 r'\cos\theta$$

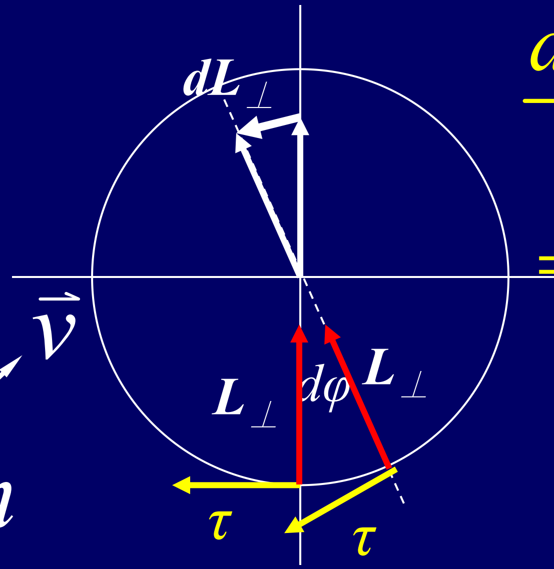
$$= m\omega^2 r^2 \sin\theta\cos\theta$$

Angular momentum and angular velocity

$p = mv$



$\tau = ?$



$$\frac{dL_\perp}{dt} = L_\perp \frac{d\varphi}{dt} = L_\perp \omega$$

$$= mr^2\omega^2 \cos\theta \sin\theta = \tau$$

$$\tau = \frac{dL_\perp}{dt}$$

$$L_\perp = r_z m v = r_z m \omega r' = (mr\cos\theta)(\omega r \sin\theta)$$

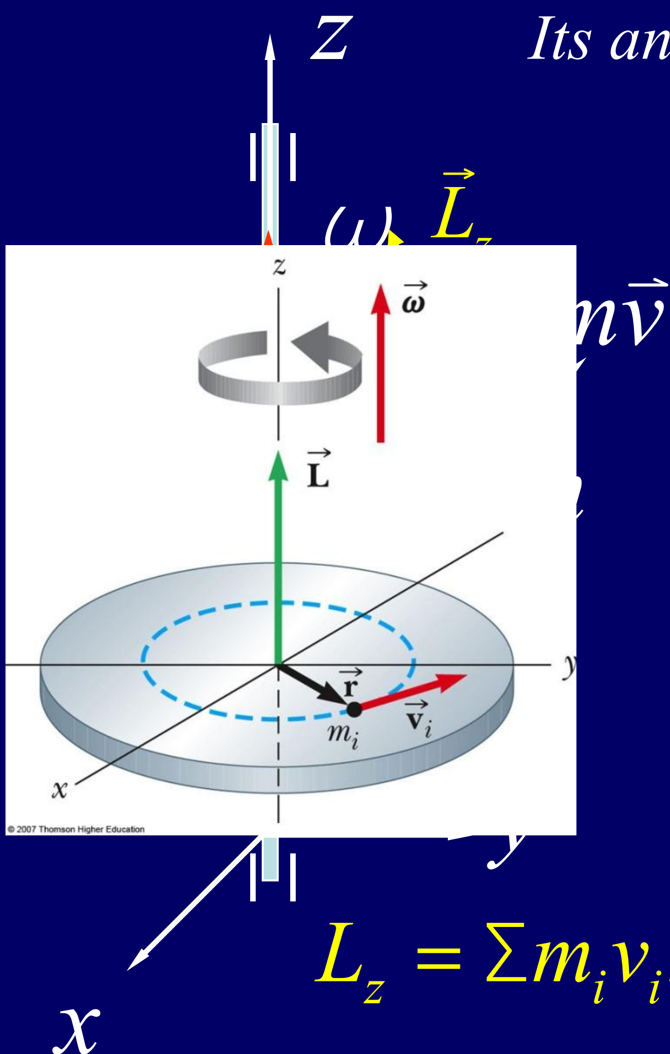
$$= mr^2\omega\cos\theta\sin\theta$$

$$\vec{\tau} = \vec{r} \times \vec{F} \qquad F = m\omega^2 r'$$

$$\tau = rF\sin(\pi/2 + \theta) = rm\omega^2 r'\cos\theta$$

$$= m\omega^2 r^2 \sin\theta\cos\theta$$

Angular momentum and angular velocity



Its angular momentum with respect to O is:

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

$$= (\vec{r}_z + \vec{r}\,') \times m\vec{v}$$

$$= \vec{r}_z \times m\vec{v} + \vec{r}\,' \times m\vec{v} = \vec{L}_\perp + \vec{L}_z$$

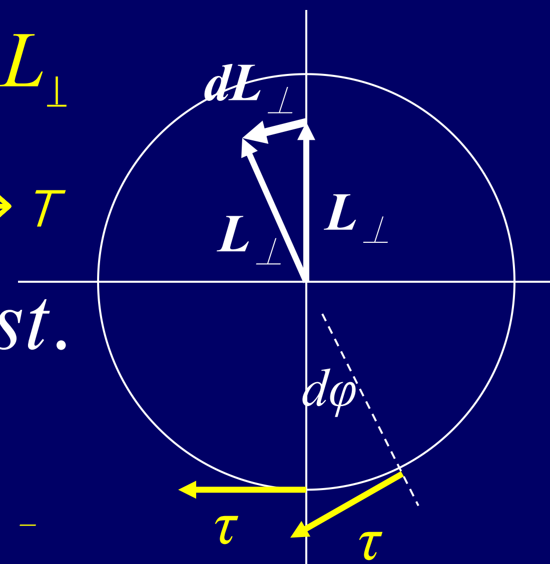
$$\frac{d\vec{L}_\perp}{dt} = \vec{\tau}$$

$$p \rightarrow L_\perp$$

$$F_c \rightarrow \tau$$

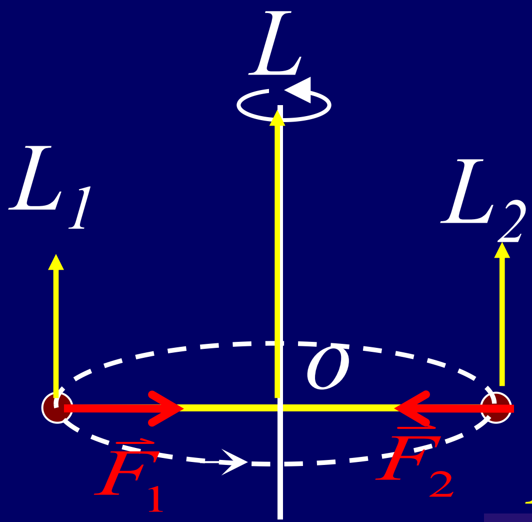
$$\vec{L}_z = I_z \vec{\omega} = const.$$

$$L_z = \sum m_i v_i r_i$$



Two particles

*If the body is **symmetric** about the rotation axis.*

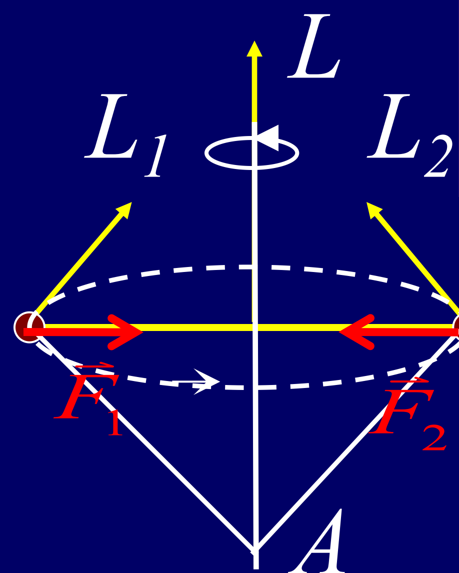


$$\vec{F} \neq const. \quad \vec{p}_1 \neq const. \quad \vec{p}_2 \neq const.$$

$$\vec{\tau}_{1,2} = \frac{d\vec{L}_{1,2}}{dt} = 0$$

$$\vec{L}_1 = const. \; \vec{L}_2 = const.$$

$$\vec{L} = \vec{L}_1 + \vec{L}_2 = const.$$



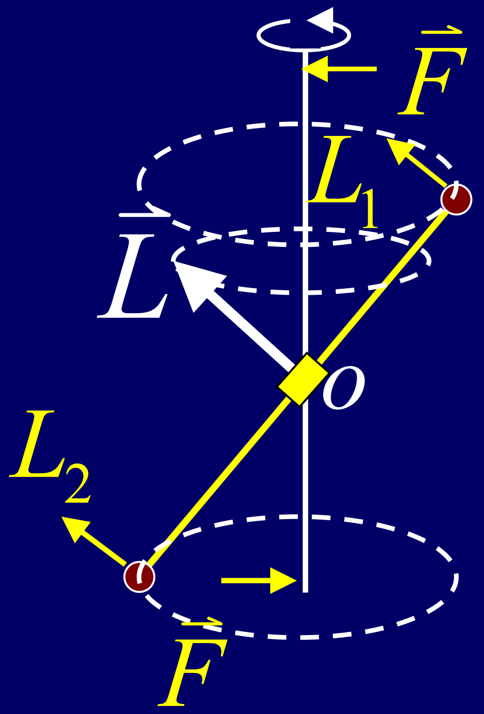
$$\vec{\tau}_{1,2} = \frac{d\vec{L}}{dt} \neq 0$$

$$\vec{L}_1 \neq const.$$

$$\vec{L}_2 \neq const.$$

$$\vec{\tau} = \vec{\tau}_1 + \vec{\tau}_2 = \frac{d\vec{L}}{dt} = 0 \quad \vec{L} = \vec{L}_1 + \vec{L}_2 = const.$$

Asymmetrical



$$\vec{\tau} = \frac{d\vec{L}}{dt} \neq 0$$

*If the body is **asymmetric, t**here is a torque acting on the axis*

Systems of particles

$$\vec{L} = \vec{L}_1 + \vec{L}_2 + \cdots + \vec{L}_N = \sum_{i=1}^{N} \vec{L}_i = \sum_{i=1}^{N} \vec{r}_i \times \vec{p}_i$$

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^{N} \frac{d\vec{L}_i}{dt} = \sum_{i=1}^{N} \frac{d}{dt}(\vec{r}_i \times \vec{p}_i) = \sum_{i=1}^{N} \left(\frac{d\vec{r}_i}{dt} \times \vec{p}_i + \vec{r}_i \times \frac{d\vec{p}_i}{dt}\right)$$

$$= \sum_{i=1}^{N} \vec{r}_i \times \vec{F}_i = \sum_{i=1}^{N} \vec{r}_i \times (\vec{f}_{i,in} + \vec{F}_{i,ext}) = \sum_{i=1}^{N} \vec{\tau}_{i,ext} = \vec{\mathrm{T}}_{ext}$$

Conservation of angular momentum

$$\sum \vec{\tau}_{ext} = \frac{d\vec{L}}{dt},\ i$$

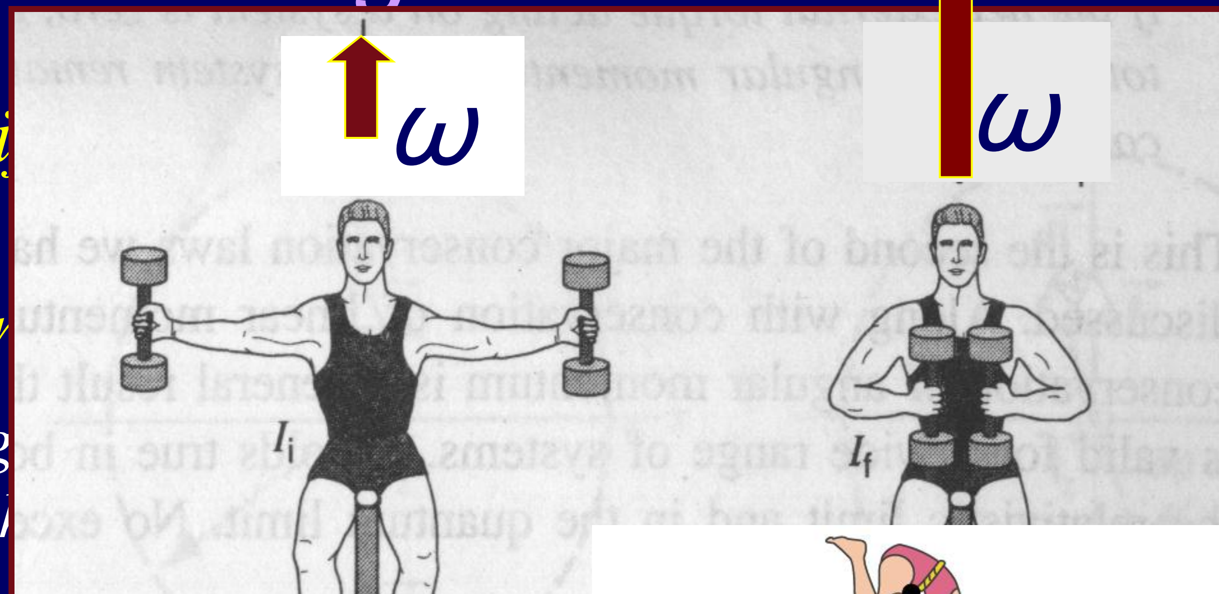
$$\vec{L} = const.\ or$$

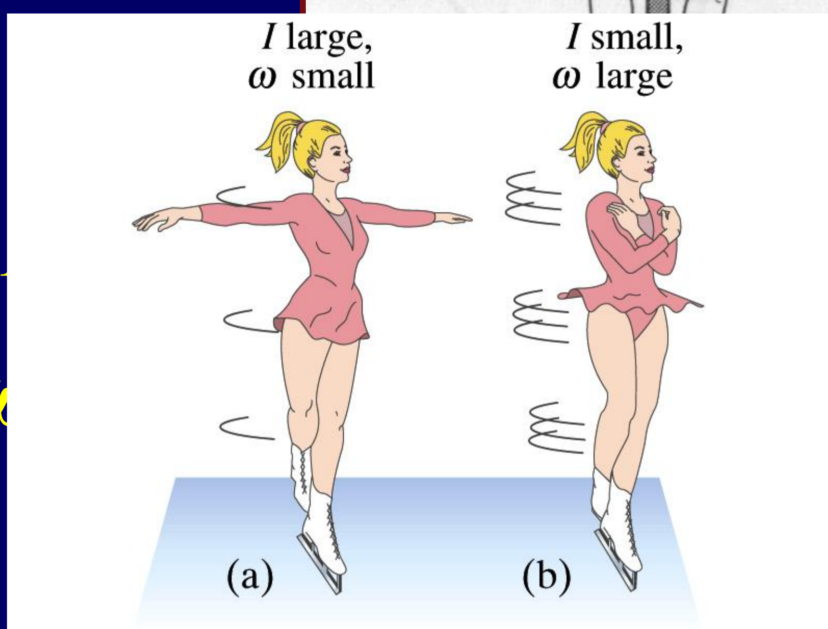
A rigid body rotating
speed ω about th

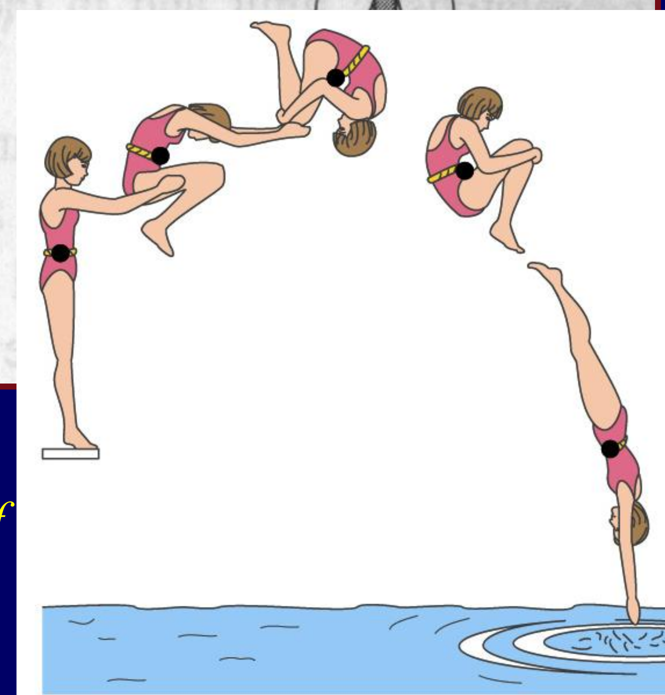
$$L_z$$

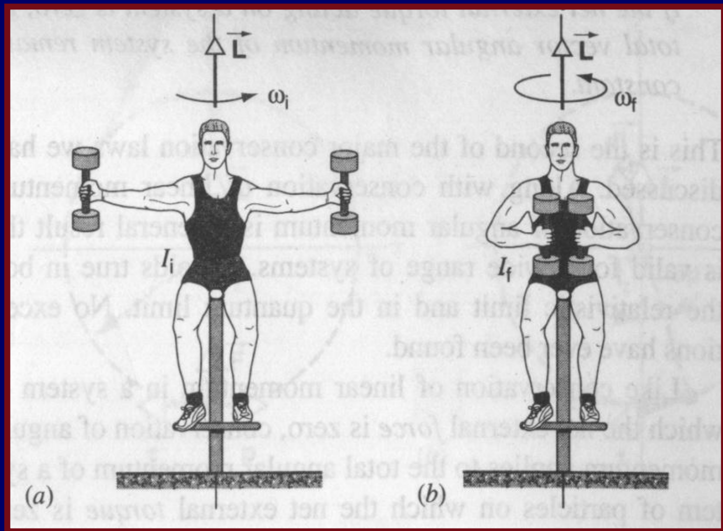
If $\tau_z=0$,

$$I_i\omega \quad f$$







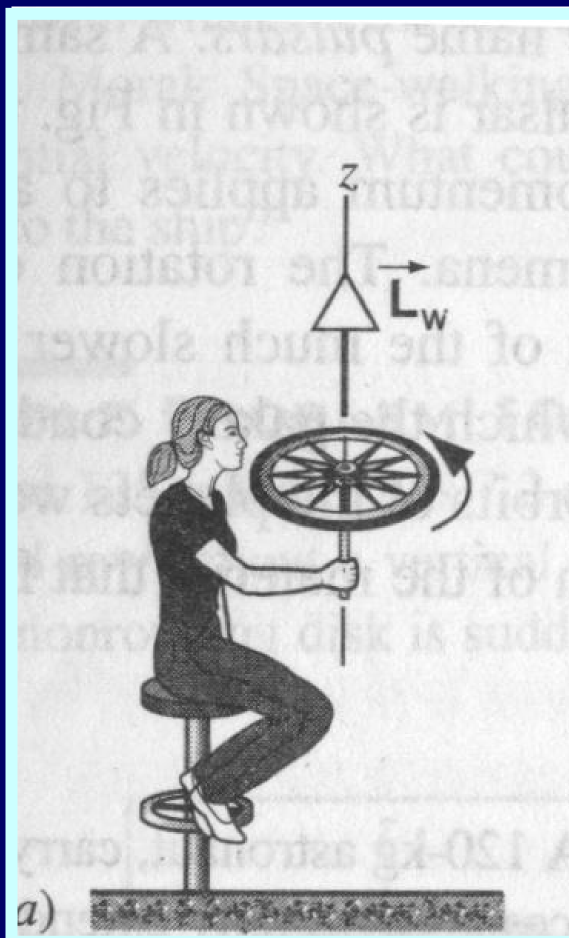


$$\omega = 20/s \qquad I = 60\times 0.15^2 + 2\times 20\times 0.5^2 = 11.35$$

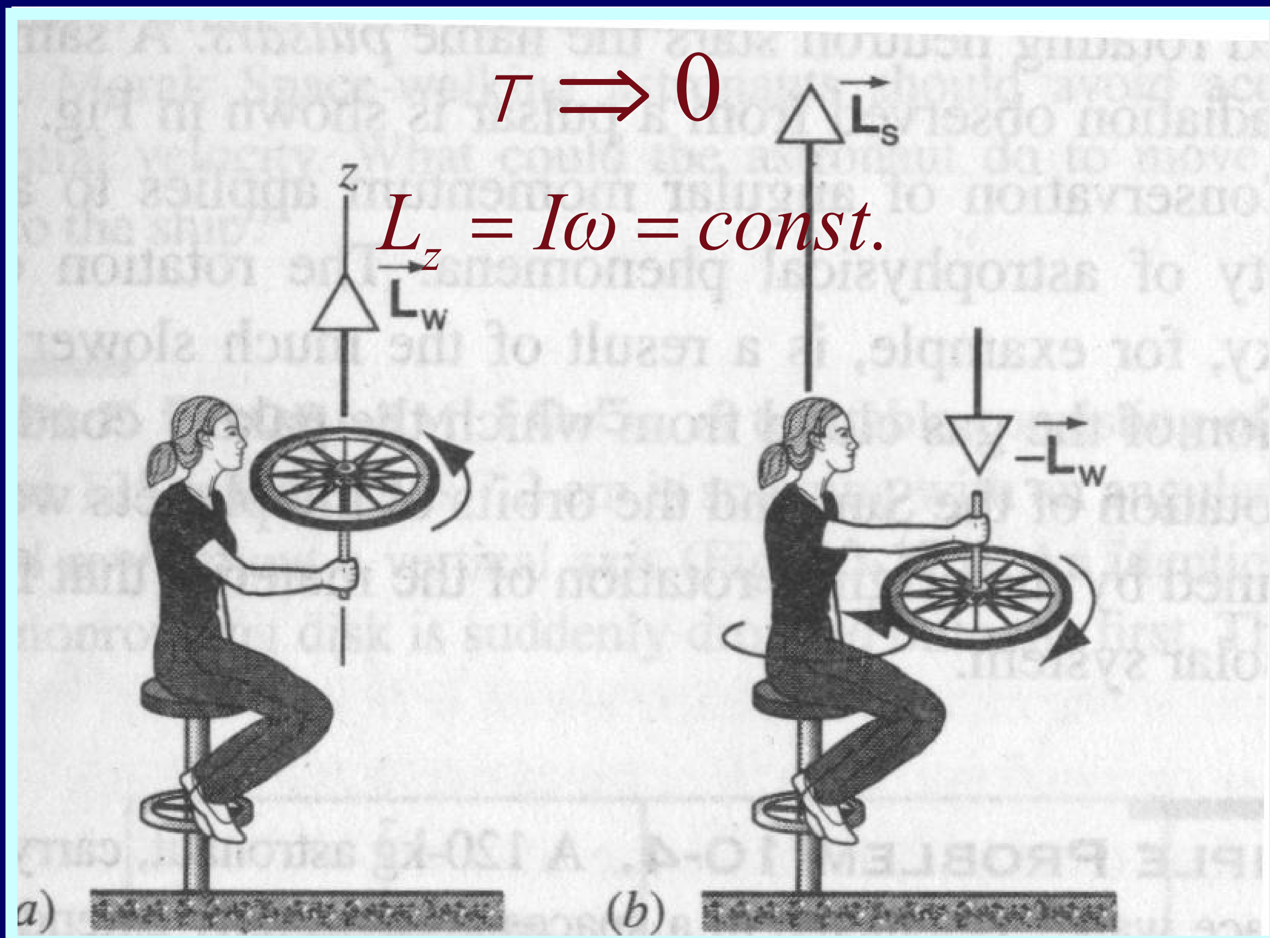
$$\omega' = 100/s \qquad I' = (60+40)\times 0.15^2 = 2.25$$

$$I\omega = I'\omega'$$

Conservation of angular momentum



Conservation of angular momentum



$$L_i = L_W$$

$$L_f = L_s - L_W$$

$$L_W = L_s - L_W$$

$$2L_W = L_s$$

Rotating about an axis

$$\tau_z = \frac{dL_z}{dt} = \frac{dI_z\omega}{dt} = I_z\frac{d\omega}{dt} = I_z\alpha$$

Translation motion

$$\Sigma F = Ma_c \qquad \tau = I\alpha$$

Rolling without slipping

$$v_{cm} = \omega R \qquad a_{cm} = \alpha R$$

About B point

Instantaneous axis

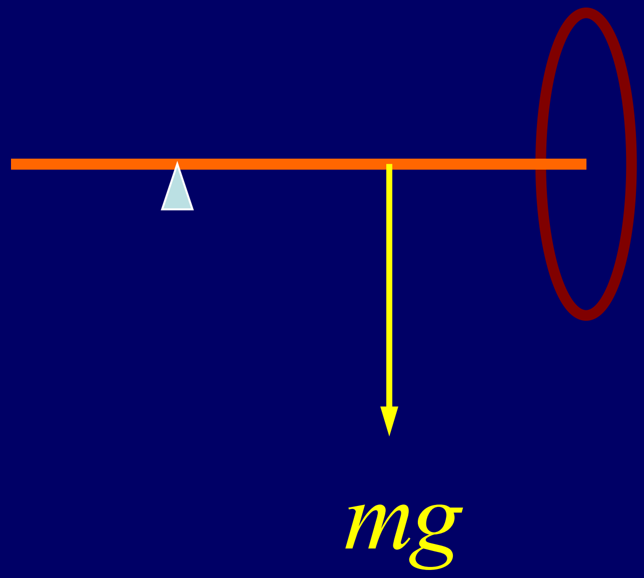
Rolling with slipping

$$v_{cm} \neq \omega R \qquad a_{cm} \neq \alpha R$$

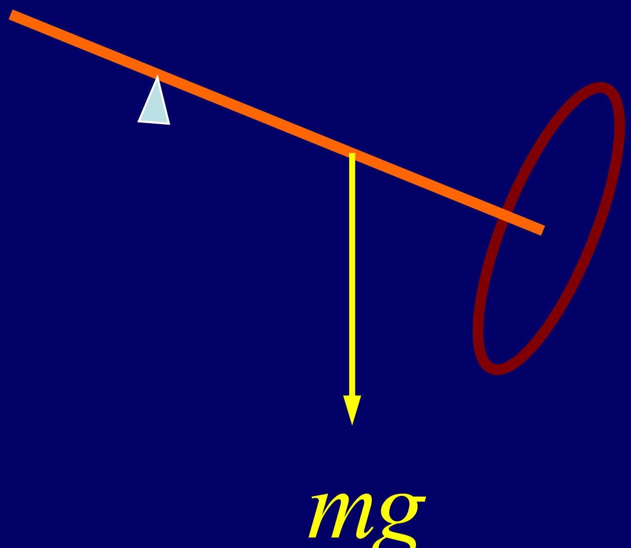
Rotating about a point

Spinning Top （陀螺）

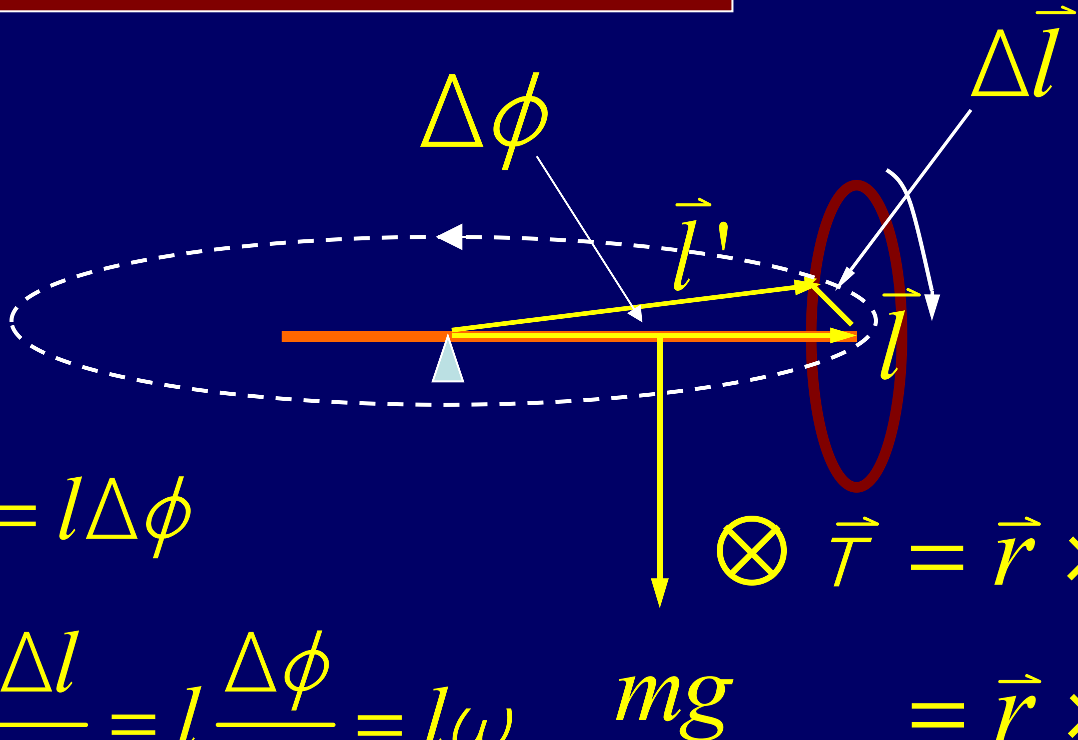
Spinning Top



Spinning Top



Spinning Top



$$\otimes\ \vec{\tau} = \vec{r} \times \vec{F}$$

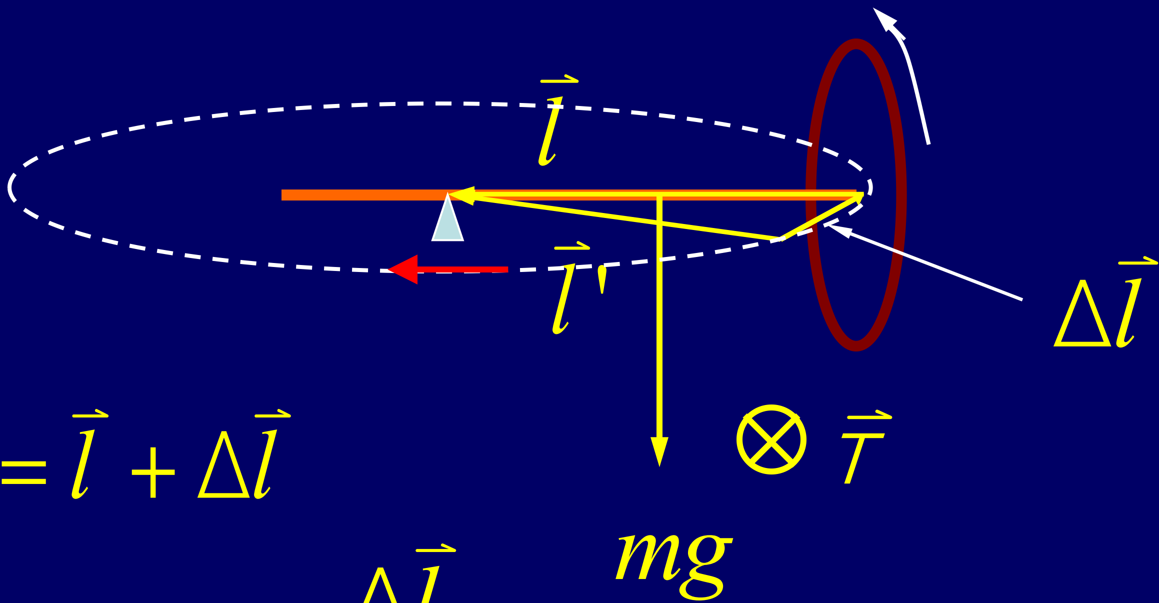
$$= \vec{r} \times m\vec{g}$$

$$\Delta l = l\Delta\phi$$

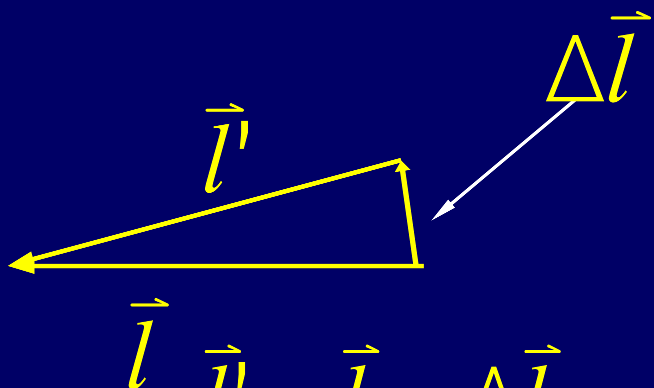
$$\tau = \frac{\Delta l}{\Delta t} = l\frac{\Delta\phi}{\Delta t} = l\omega_p$$

$$\omega_p = \frac{\tau}{l}$$

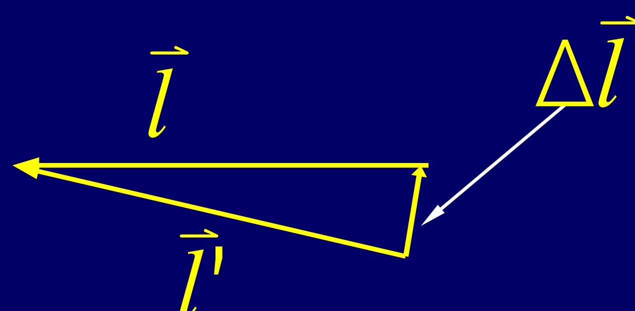
Spinning Top



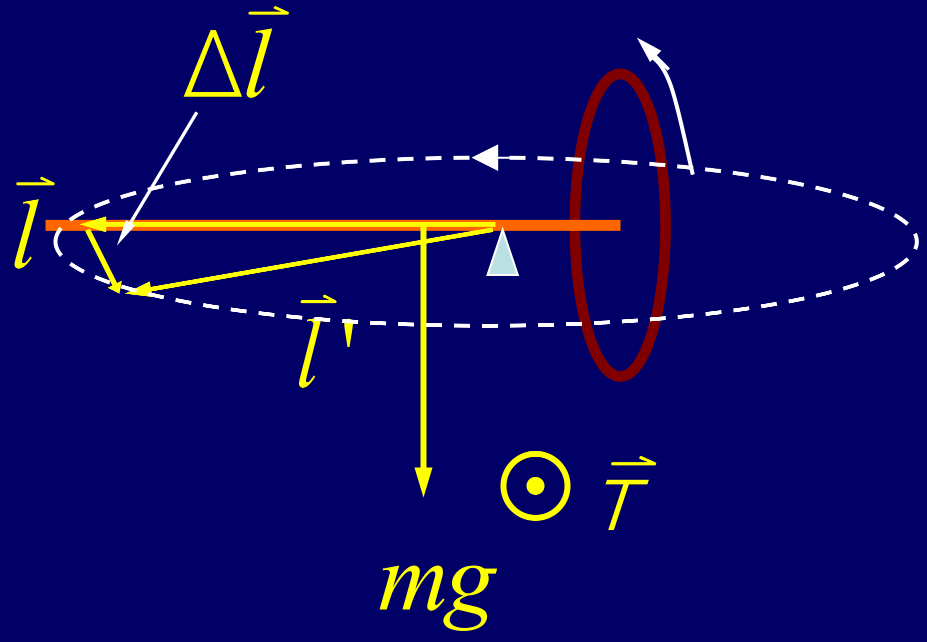
$$\vec{l}\,' = \vec{l} + \Delta\vec{l}$$

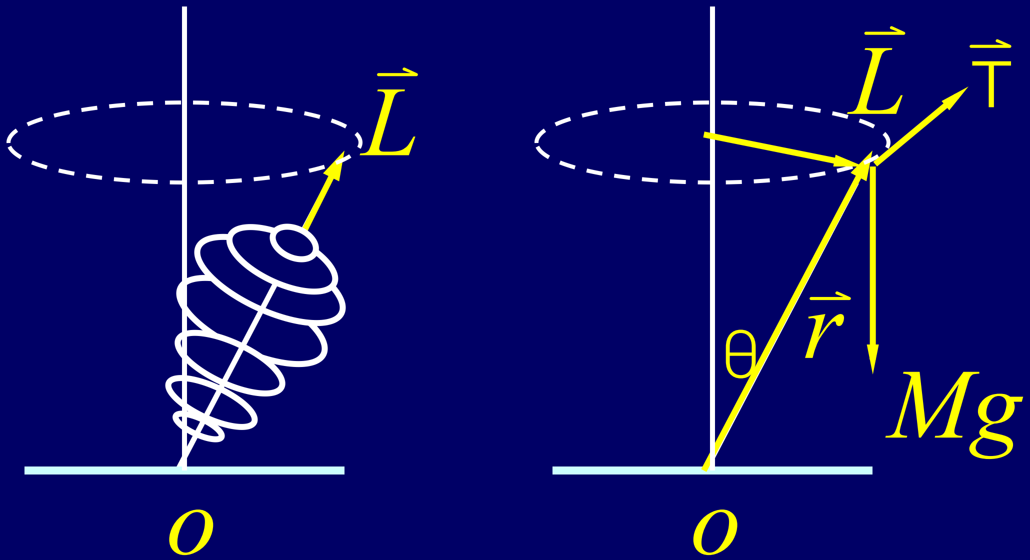


$$\vec{l}\,' = \vec{l} - \Delta\vec{l}$$

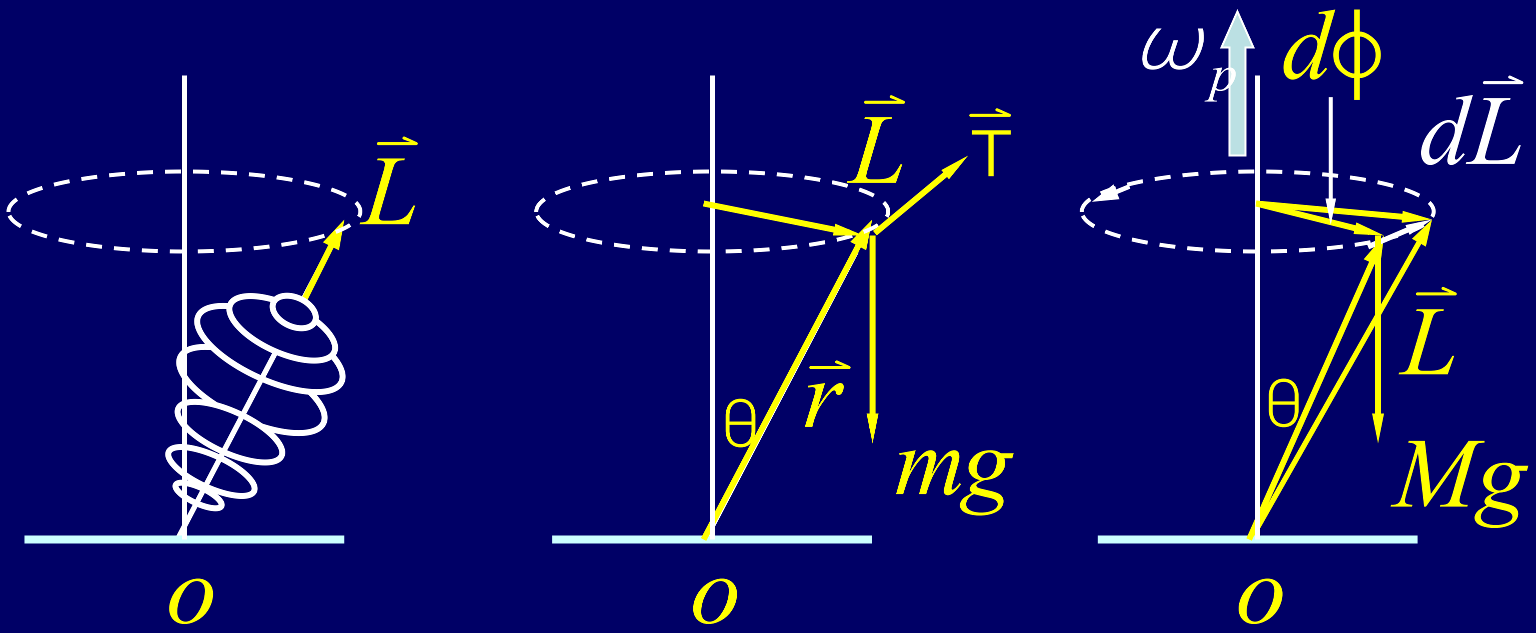


Spinning Top





$$\tau = Mgr\sin\theta; \quad and \;\; \vec{\tau} \perp \vec{L}$$



$$\tau = Mgr\sin\theta = \omega_p L\sin\theta \qquad \vec{\tau} = \vec{\omega}_P \times \vec{L}$$

$$\because \vec{\tau} = \frac{d\vec{L}}{dt}; \qquad \therefore d\vec{L} = \vec{L}(t+dt) - \vec{L}(t) = \vec{\tau}dt$$

$$L\sin\theta d\phi = dL = (\,Mgr\sin\theta\,)dt$$

$$\omega_p = \frac{d\phi}{dt} = \frac{(Mgr\sin\theta)}{L\sin\theta} = \frac{Mgr}{L}$$

Spinning Top

Review of Rotational Dynamics

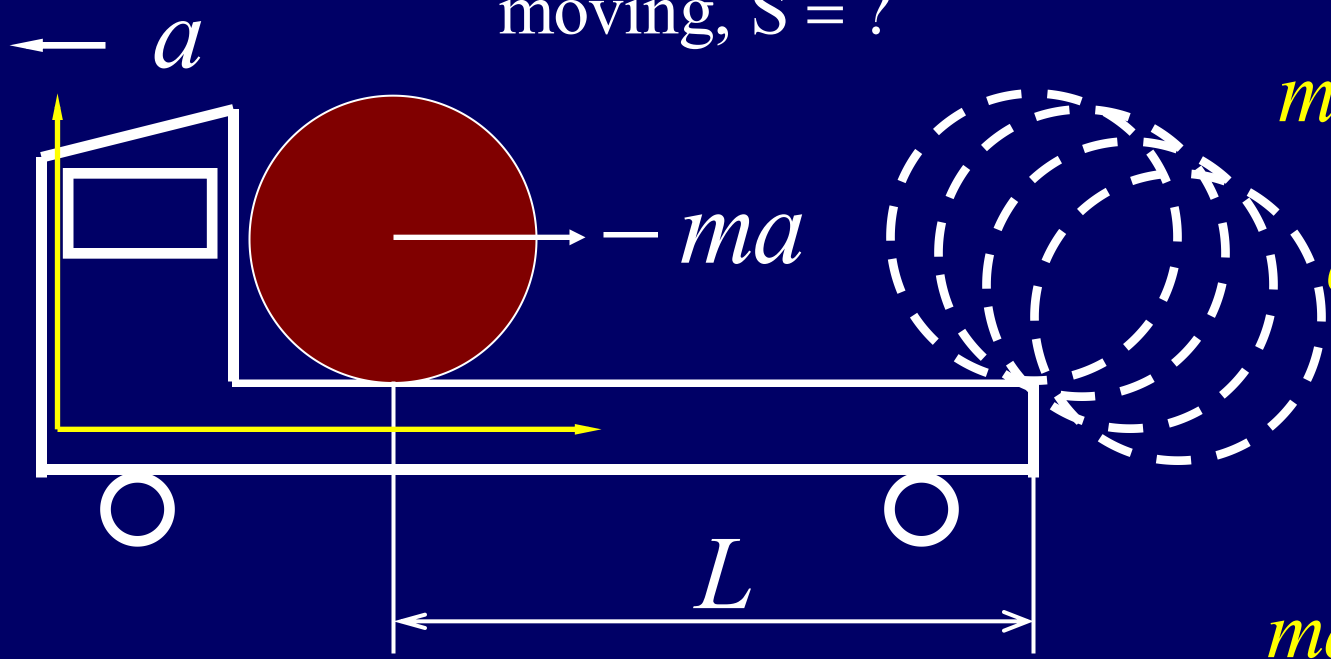
	Translational Quantity	Rotational Quantity
Mass	m	$I = \sum mr^2$
Force	$\vec{F}$	$\vec{\tau} = \vec{r} \times \vec{F}$
Velocity	$\vec{v} = \dfrac{d\vec{r}}{dt}$	$\vec{\omega} = \dfrac{\overrightarrow{d\varphi}}{dt}$
Acceleration	$\vec{a} = \dfrac{d\vec{v}}{dt}$	$\vec{\alpha} = \dfrac{d\vec{\omega}}{dt}$

Review of Rotational Dynamics

	Translational Quantity	Rotational Quantity
Momentum	$\vec{P} = m\vec{v}$	$\vec{l} = \vec{r} \times \vec{p}$
Momentum of system	$\vec{P} = M\vec{v}_{cm}$	$\vec{L}_z = I_z\vec{\omega}$
Newton's law	$\sum\vec{F}_{ex} = m\vec{a}_{cm}$	$\sum\vec{\tau}_{ex} = I\vec{\alpha}$
Equilibrium condition	$\sum\vec{F}_{ex} = \frac{d\vec{p}}{dt} = 0$	$\sum\vec{\tau}_{ex} = \frac{d\vec{L}}{dt} = 0$
	$\sum\vec{p} = const.$	$\sum\vec{l} = const.$

Example

How far is the truck moving, S = ?



$$fR = (\frac{1}{2}mR^2)\alpha$$

$$ma - f = ma_{cm}$$

$$a - \frac{1}{2}R\alpha = a_{cm}$$

$$\alpha = \frac{2a}{3R}$$

$$maR = (\frac{3}{2}mR^2)\alpha$$

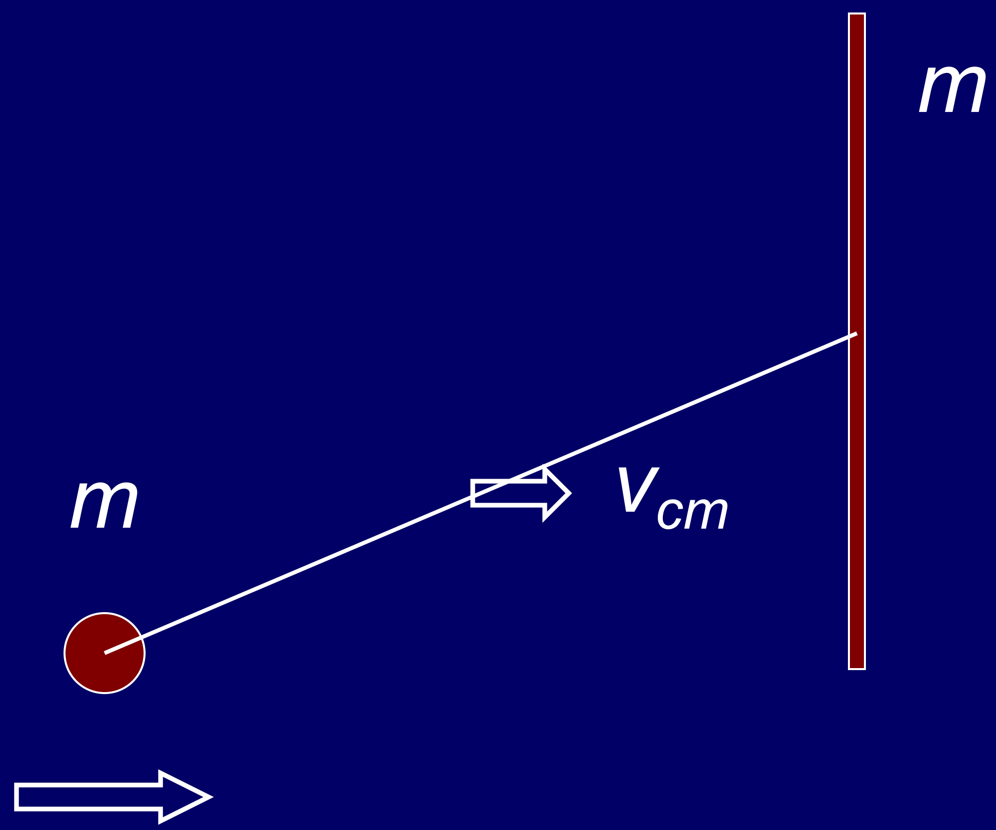
$$R\alpha = \frac{2}{3}a$$

$$a_{cm} = \frac{2}{3}a \qquad L = \frac{1}{2}a_{cm}t^2$$

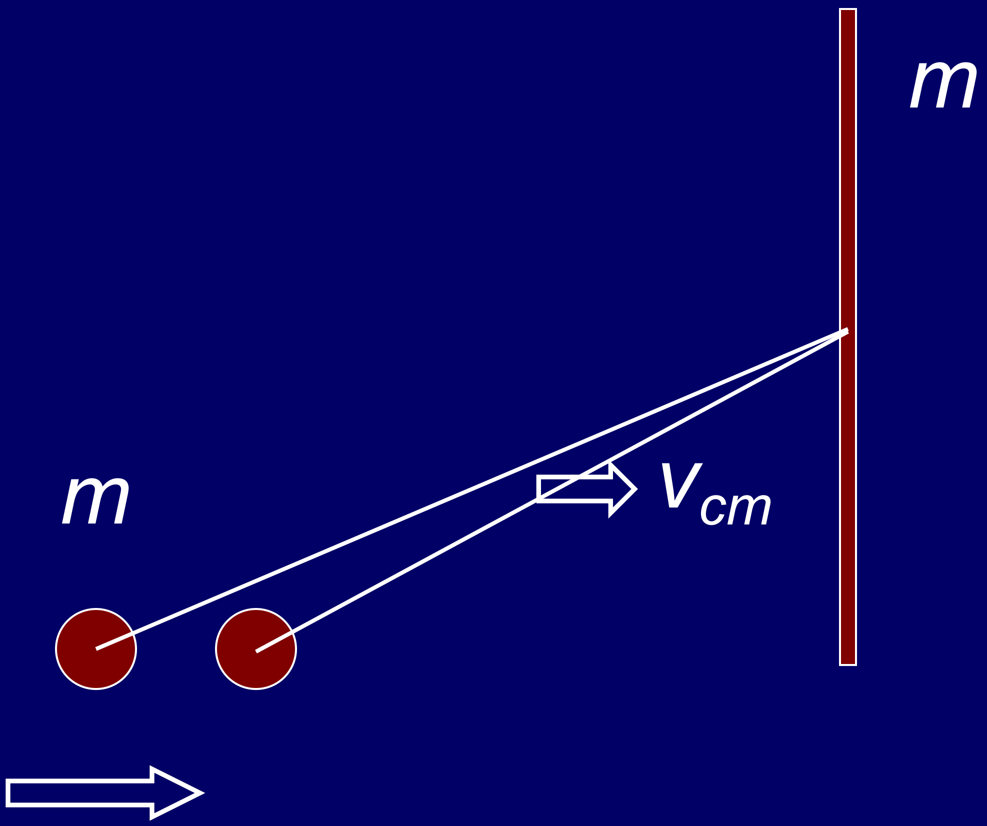
$$t = \sqrt{\frac{2L}{a_{cm}}} = \sqrt{\frac{3L}{a}}$$

$$S = \frac{1}{2}at^2 = \frac{3}{2}L$$

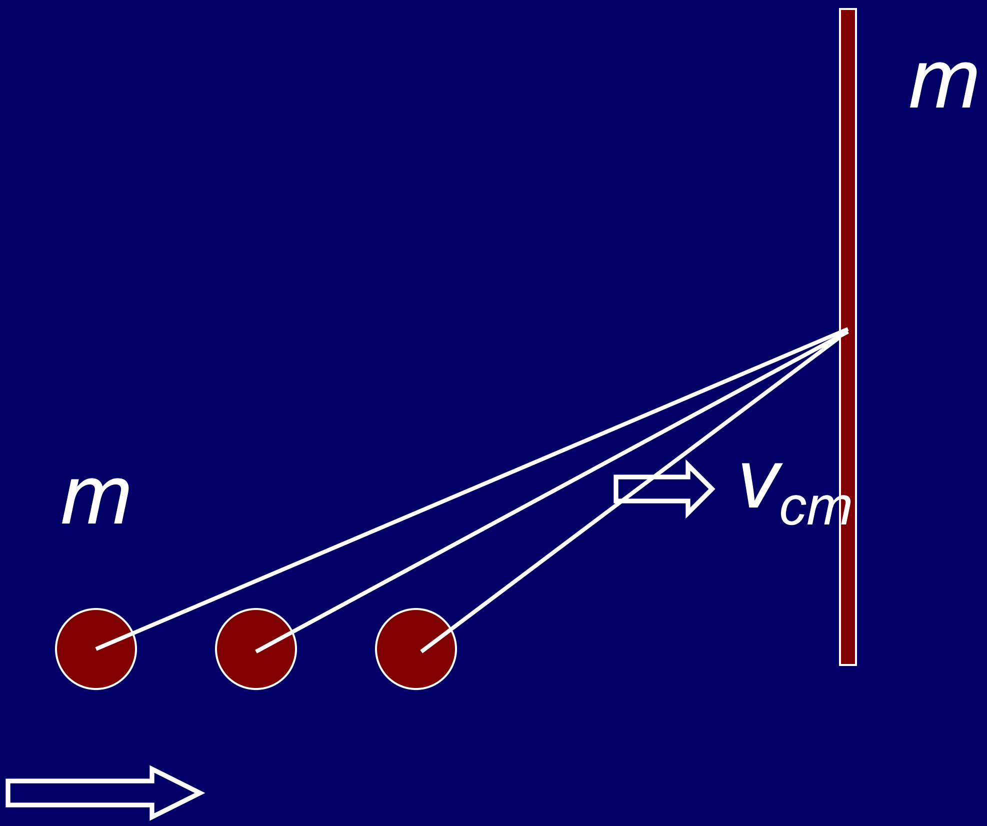
Example



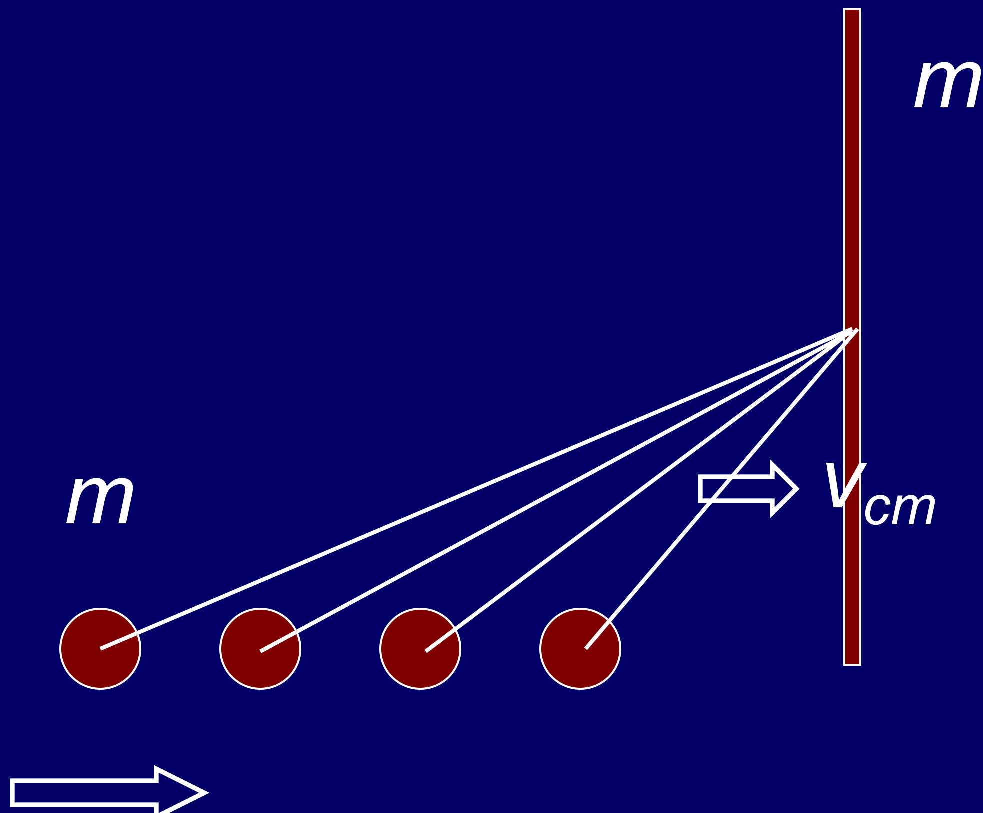
Example



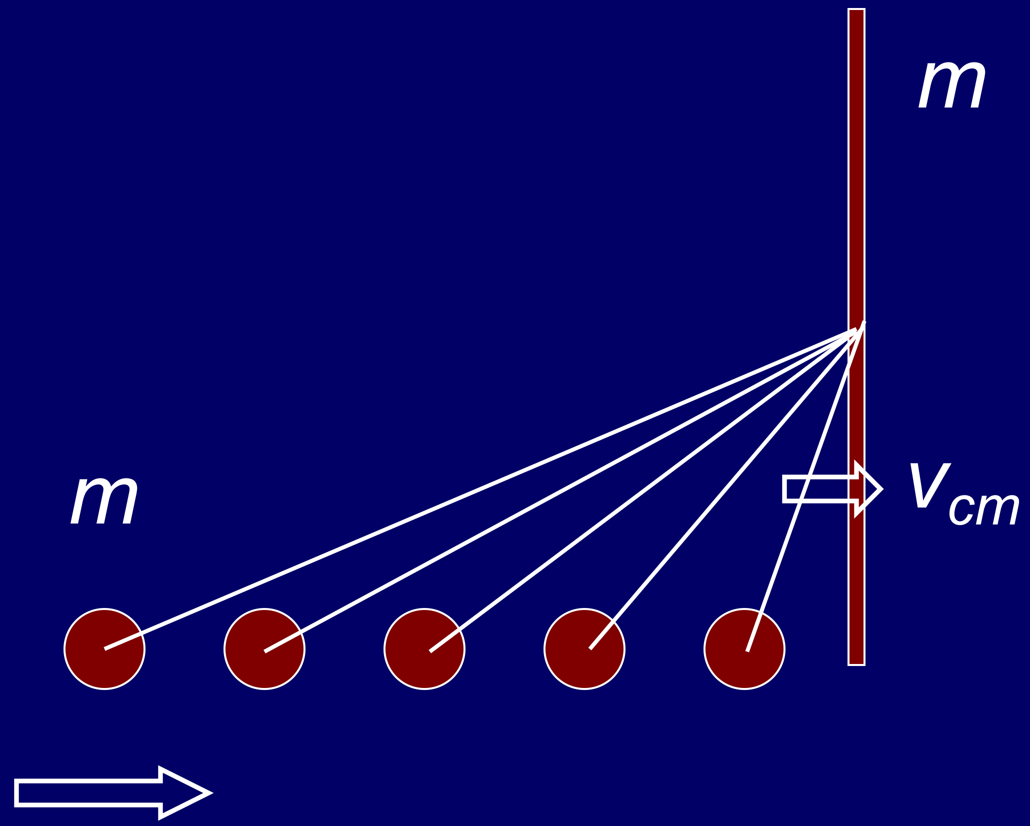
Example



Example



Example

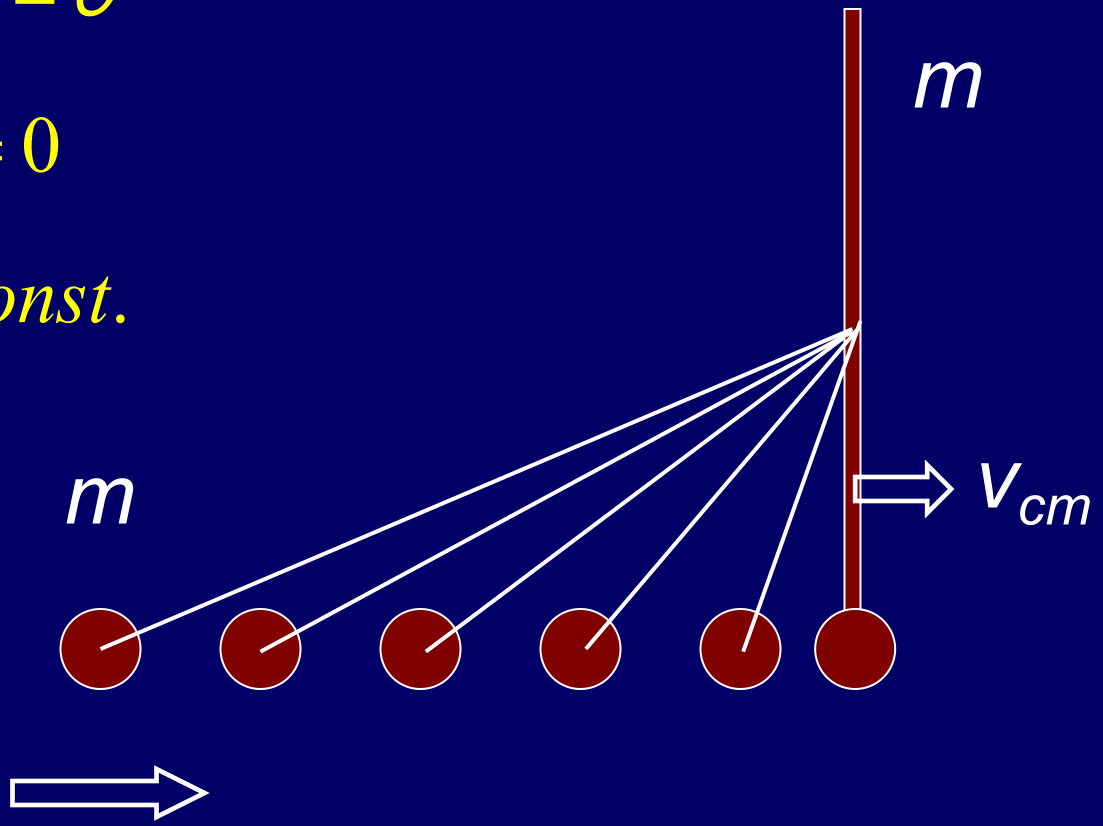


Example

$$\sum \vec{F}_{ext} = 0$$

$$\vec{a}_{cm} = 0$$

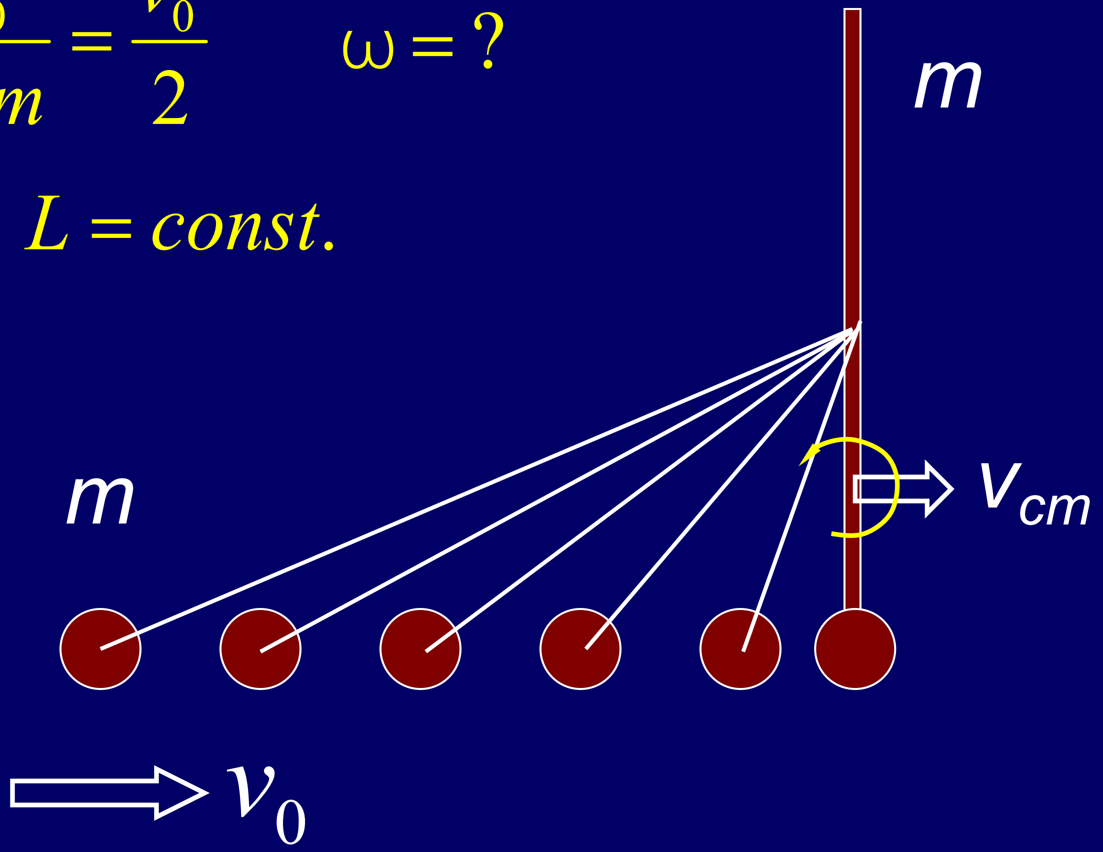
$$\vec{v}_{cm} = const.$$



Example

$$\vec{v}_{cm} = \frac{mv_0}{m+m} = \frac{v_0}{2} \qquad \omega = ?$$

$$\tau_{ex} = 0 \qquad L = const.$$



$$L_i = \vec{l}_{1i} = \vec{r} \times m\vec{v}_{1i} = mrv_{1i}\sin\theta = \frac{\lambda}{4} m v_0$$ about C point

$$L_f = I\omega$$

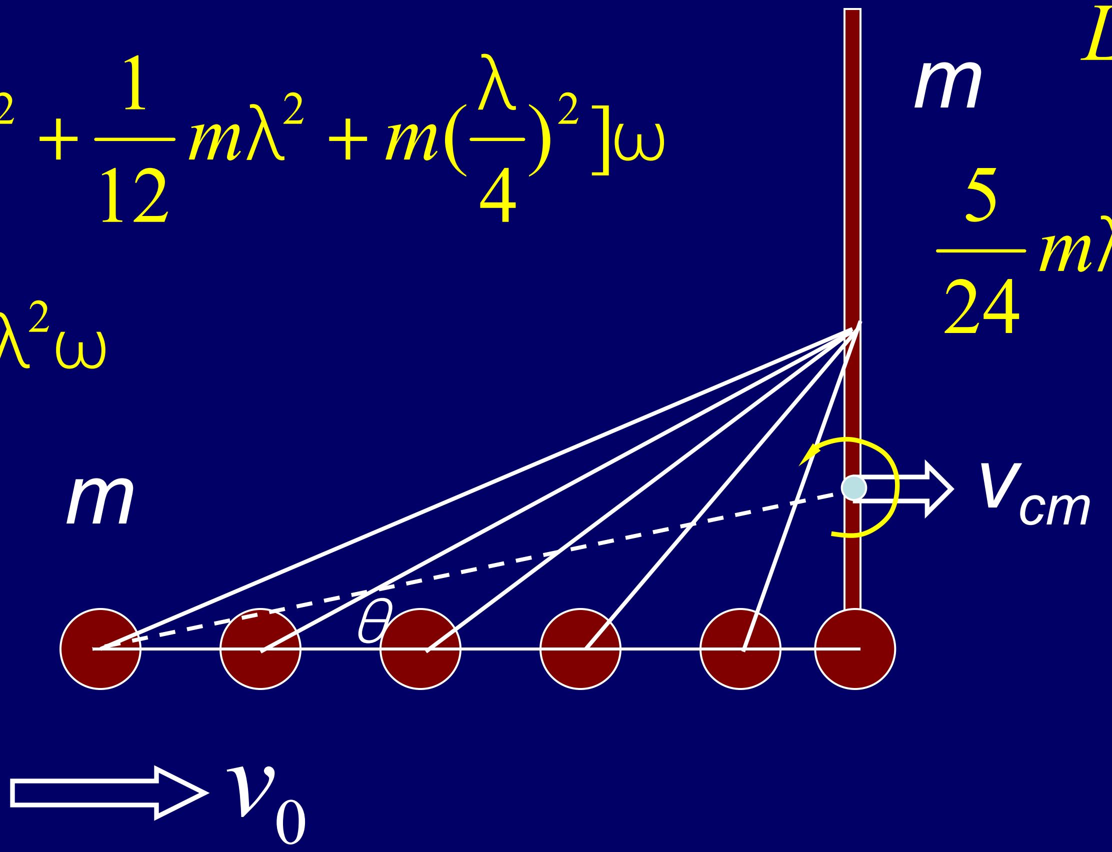
$$= [m(\frac{\lambda}{4})^2 + \frac{1}{12} m\lambda^2 + m(\frac{\lambda}{4})^2]\omega$$

$$= \frac{5}{24} m\lambda^2\omega$$

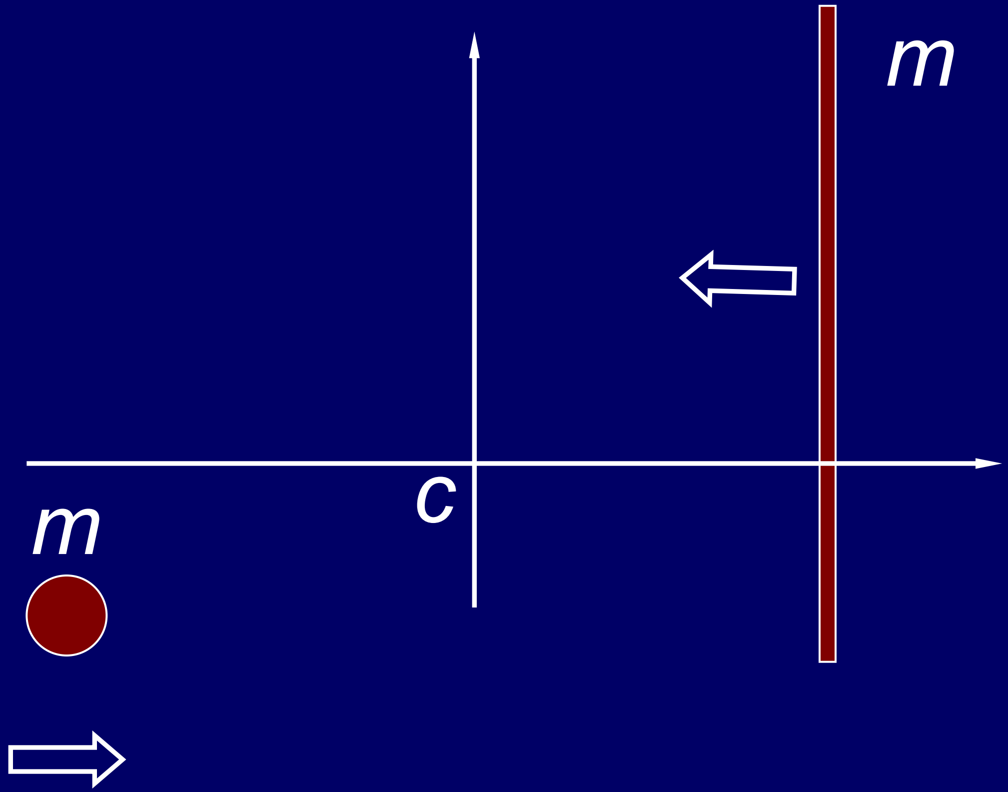
$$L = const.$$

$$\frac{5}{24} m\lambda^2\omega = \frac{\lambda}{4} m v_0$$

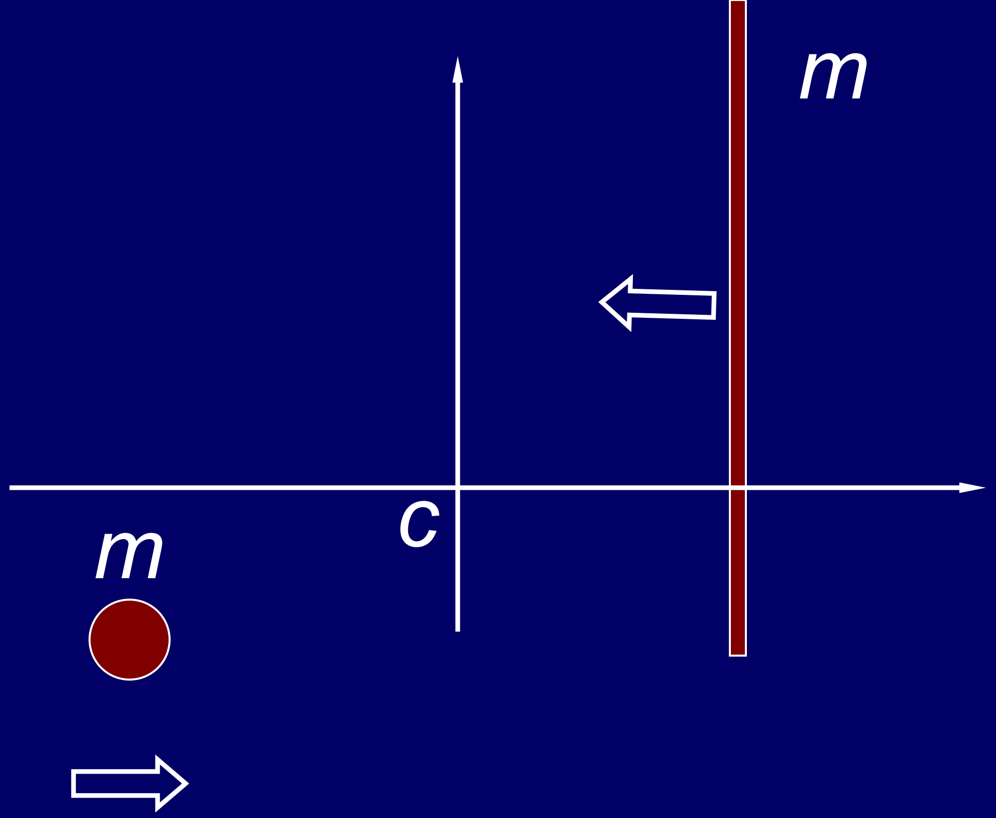
$$\omega = \frac{6v_0}{5\lambda}$$



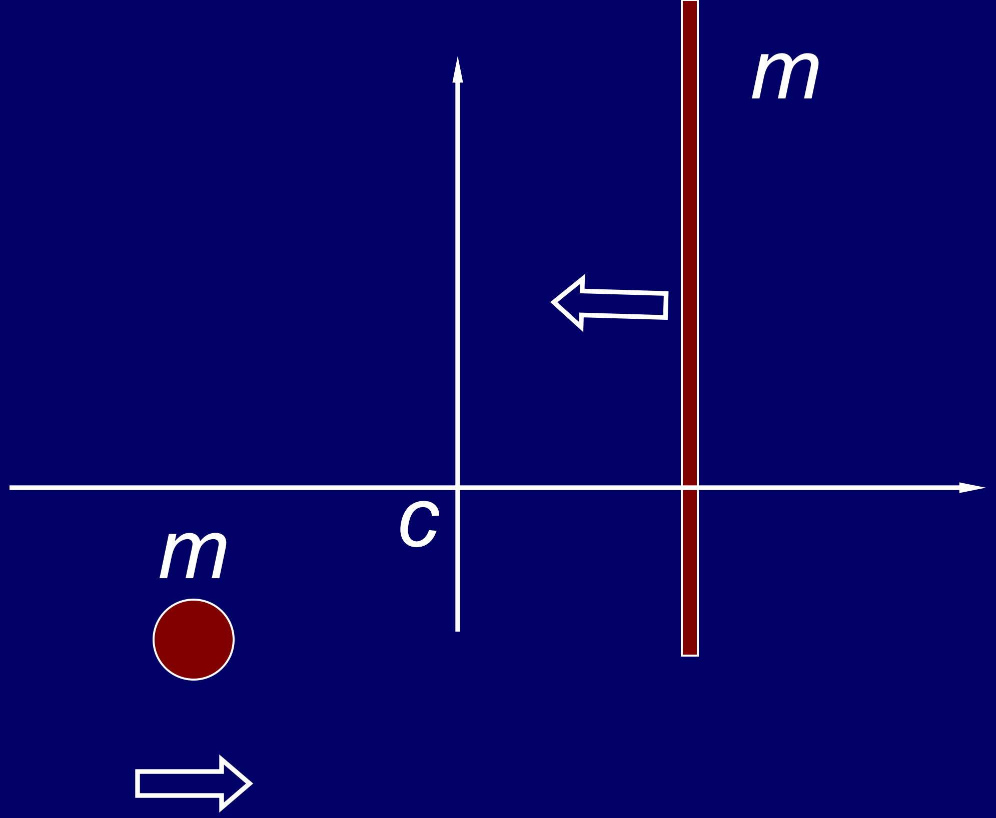
cm Frame



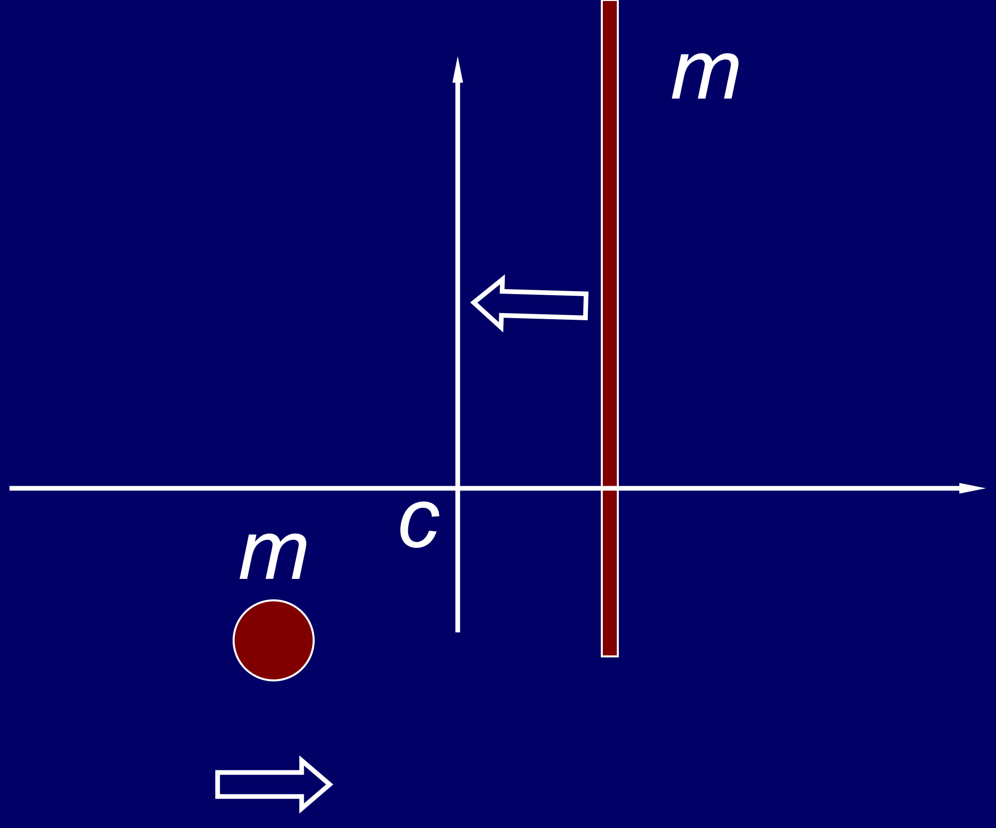
cm Frame



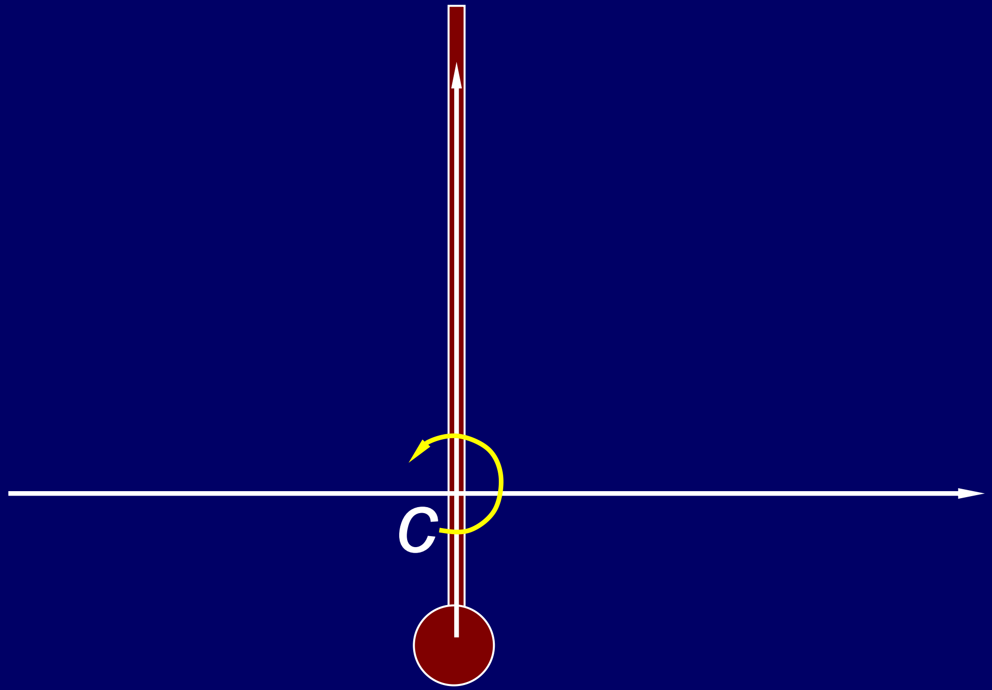
cm Frame



cm Frame

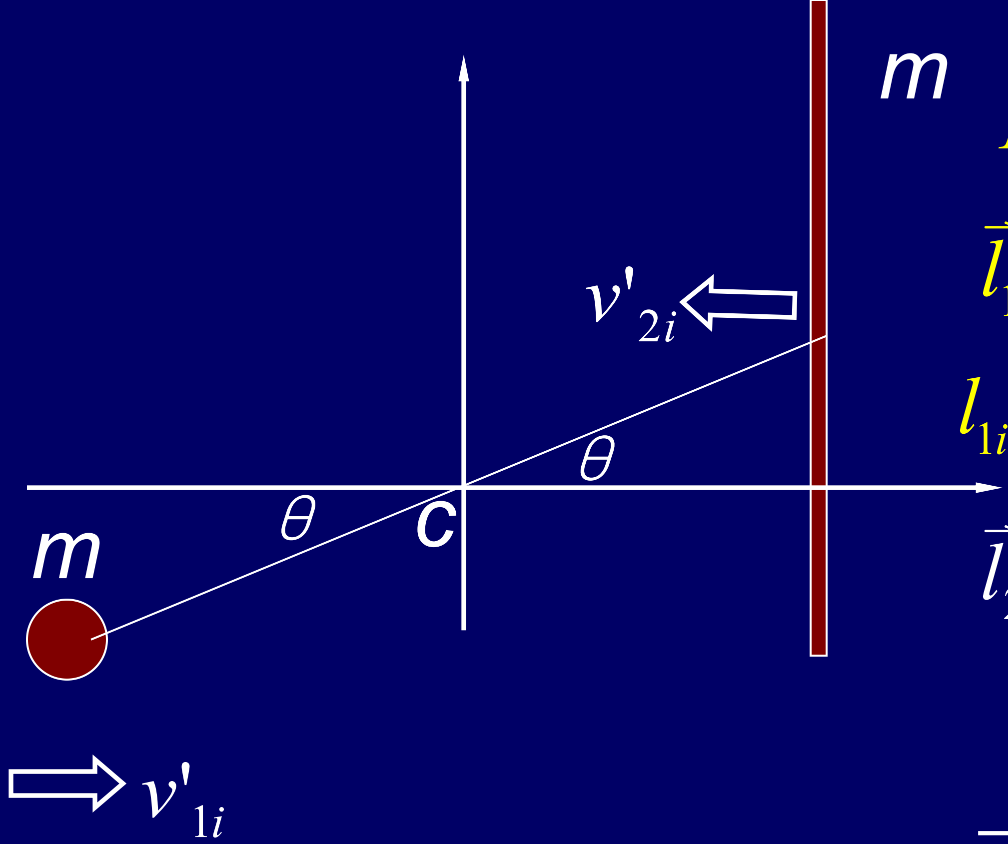


cm Frame



$$\vec{v}'_{cm} = 0$$

$$\omega = ?$$



$$\vec{L}_i = \vec{l}_{1i} + \vec{l}_{2i}$$

$$\vec{l}_{1i} = \vec{r}'_{1i} \times m\vec{v}'_{1i}$$

$$l_{1i} = mr'_{1i}\, v'_{1i} \sin\theta = m\frac{\lambda}{4}\frac{v_0}{2}$$

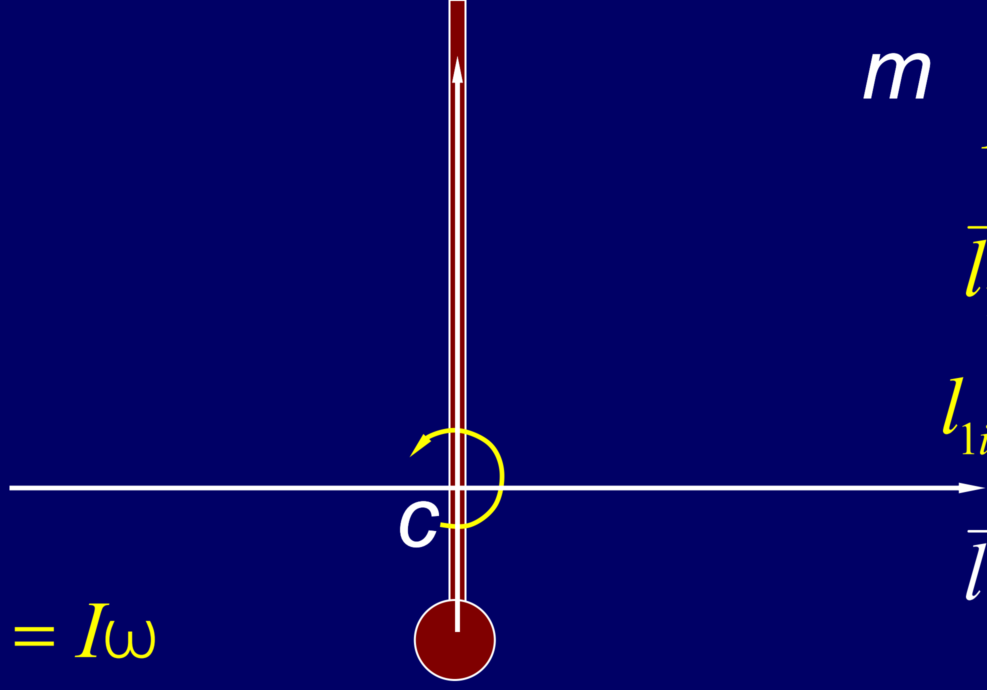
$$\vec{l}_{2i} = \sum \vec{r}'_n \times m_n \vec{v}'_{2i}$$

$$= \left(\sum m_n \vec{r}'_n\right) \times \vec{v}'_{2i}$$

$$= m\vec{r}'_{cm} \times \vec{v}'_{2i} = \vec{r}'_{cm} \times m\vec{v}'_{2i}$$

$$l_{2i} = mr'_{cm}\, v'_{2i} \sin\theta = m\frac{\lambda}{4}\frac{v_0}{2}$$

$$L_i = m\frac{\lambda}{4}\frac{v_0}{2} + m\frac{\lambda}{4}\frac{v_0}{2} = \frac{\lambda}{4} m v_0$$



$$\vec{L}_i = \vec{l}_{1i} + \vec{l}_{2i}$$

$$\vec{l}_{1i} = \vec{r}\,'_{1i} \times m\vec{v}\,'_{1i}$$

$$l_{1i} = mr'_{1i}\, v'_{1i} \sin\theta = m\frac{\lambda}{4}\frac{v_0}{2}$$

$$\vec{l}_{2i} = \sum \vec{r}\,'_n \times m_n \vec{v}\,'_{2i}$$

$$= \left(\sum m_n \vec{r}\,'_n\right) \times \vec{v}\,'_{2i}$$

$$= m\vec{r}\,'_{cm} \times \vec{v}\,'_{2i}$$

$$l_{2i} = mr'_{cm}\, v'_{2i} \sin\theta = m\frac{\lambda}{4}\frac{v_0}{2}$$

$$L_i = m\frac{\lambda}{4}\frac{v_0}{2} + m\frac{\lambda}{4}\frac{v_0}{2} = \frac{\lambda}{4} m v_0$$

$$L_f = I\omega$$

$$= [m(\frac{\lambda}{4})^2 + \frac{1}{12}m\lambda^2 + m(\frac{\lambda}{4})^2]\omega$$

$$= \frac{5}{24} m\lambda^2\omega$$

$$\frac{5}{24} m\lambda^2\omega = \frac{\lambda}{4} m v_0 \qquad \omega = \frac{6v_0}{5\lambda}$$

One particle

$$\vec{F} = m\vec{a} \qquad \vec{\tau} = I\vec{\alpha}$$

$$\vec{\tau} = \vec{r} \times \vec{F} \qquad I = mr^2$$

Many particles

Condition of Equilibrium

$$\Sigma\vec{F}_{ext} = M\vec{a}_{cm} = 0 \qquad \Sigma\vec{\tau}_{ext} = I\vec{\alpha} = 0$$

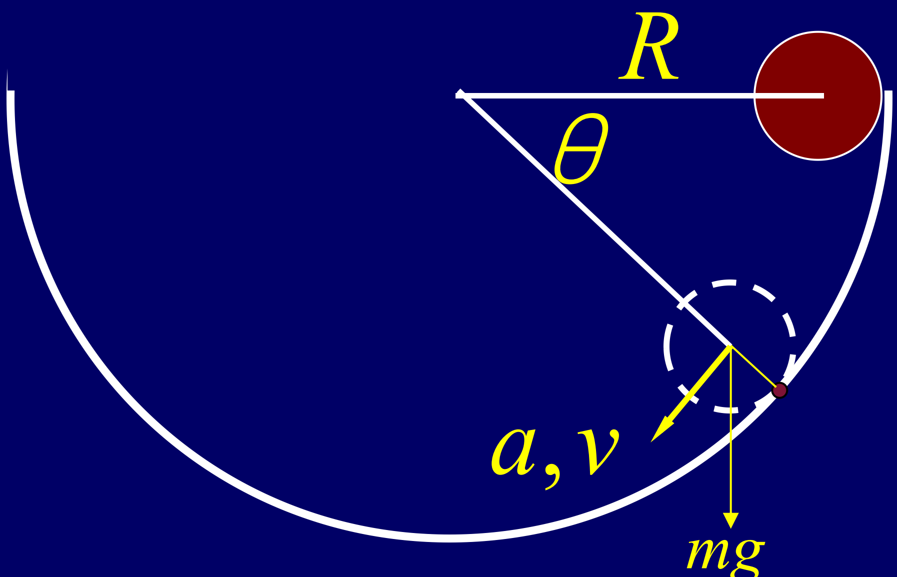
$$\Sigma\vec{\tau}_{ext} = \Sigma\vec{r}_i \times \vec{F}_{ext,i} \qquad I = \Sigma m_i r_i^2$$

Impulse

Angular Impulse

$$\int \Sigma\vec{F}_{ext}dt = \Delta\vec{p} \qquad \int \Sigma\vec{\tau}_{ext}dt = \Delta\vec{L}$$

Example



A solid ball with mass m and radius r

r<<R

Rolling without slipping

$$mgr\cos\theta = \frac{7}{5}mr^2 a$$

$$a = \frac{5g\cos\theta}{7r}$$

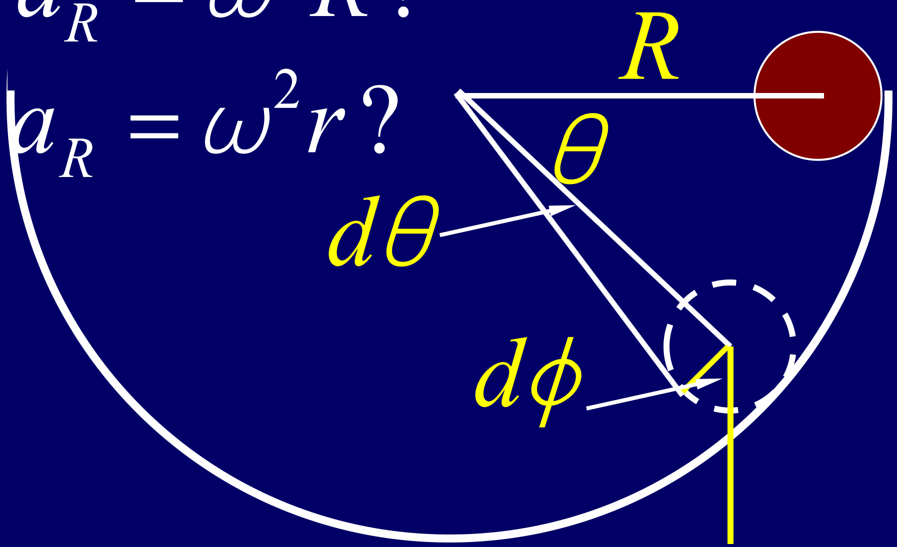
$$ar = \frac{5g\cos\theta}{7}$$

Example

$$\frac{d\theta}{dt} \neq \omega \qquad \frac{d\omega}{d\theta}\frac{d\theta}{dt} = \frac{d\omega}{dt} = \frac{5g\cos\theta}{7r}$$

$$a_R = \omega^2 R\,?$$

$$a_R = \omega^2 r\,?$$



$$\frac{d\phi}{dt} = \omega \qquad \frac{d\theta}{dt} = \frac{r}{R}\frac{d\phi}{dt}$$

$$\frac{d\omega}{d\theta}\frac{r}{R}\frac{d\phi}{dt} = \frac{5g\cos\theta}{7r}$$

$$\int_0^{\omega} \omega\, d\omega = \int_0^{\theta} \frac{5gR\cos\theta\, d\theta}{7r^2}$$

$$v = r\omega = \sqrt{\frac{10gR\sin\theta}{7}} \qquad \omega = \sqrt{\frac{10gR\sin\theta}{7r^2}}$$

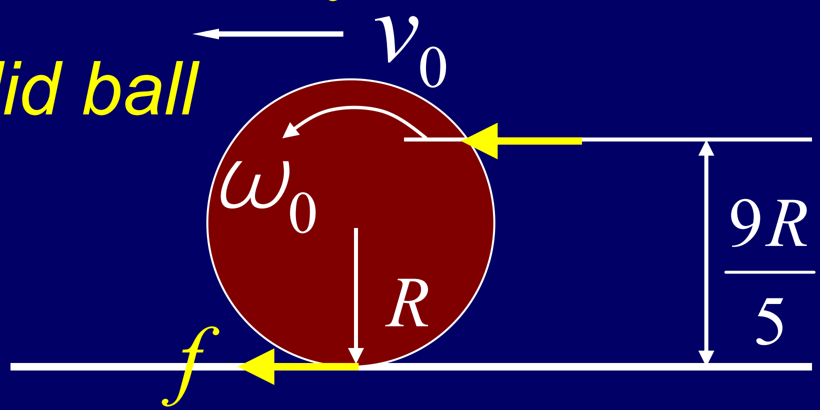
$$a_R = \frac{v^2}{R} = \frac{10g\sin\theta}{7}$$

Example

Rolling without slipping

$$\omega_0 = ? \quad v = ?$$

Solid ball



$$a = \frac{f}{m} \qquad v = v_0 + \frac{f}{m}t \qquad \frac{2}{5}v = \frac{4}{5}v_0 - \frac{f}{m}t$$

$$a = -\frac{fR}{\frac{2}{5}mR^2} \qquad v = \frac{9}{7}v_0 > v_0$$

$$F >> f$$

Impulse

$$\int F dt = mv_0$$

Angular Impulse

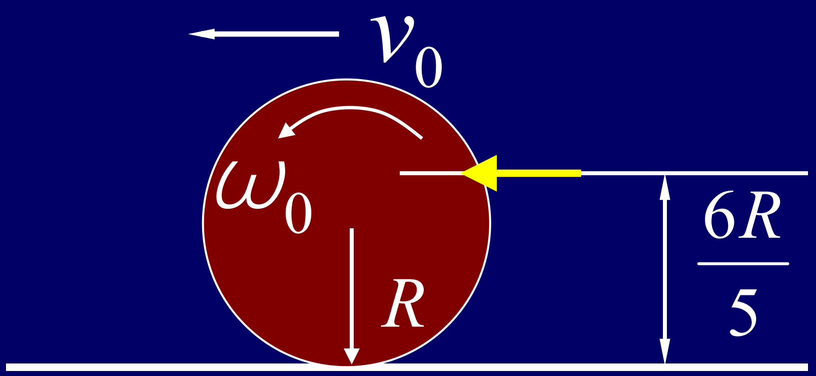
$$\int F\frac{4R}{5}dt = \frac{2}{5}mR^2\omega_0$$

$$2\int F dt = mR\omega_0$$

$$2mv_0 = mR\omega_0$$

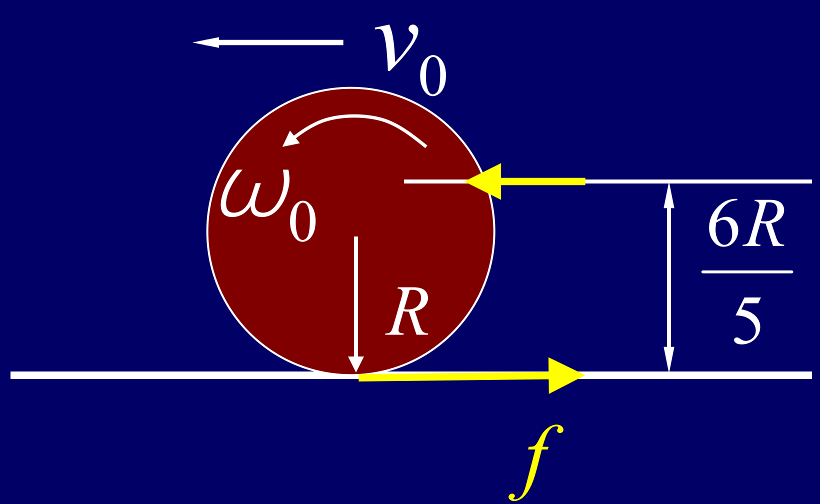
$$\omega_0 = \frac{2v_0}{R}$$

Example



$$v = ?$$

Example



$$2mR\omega_0 = mv_0$$

$$2R\omega_0 = v_0$$

Impulse

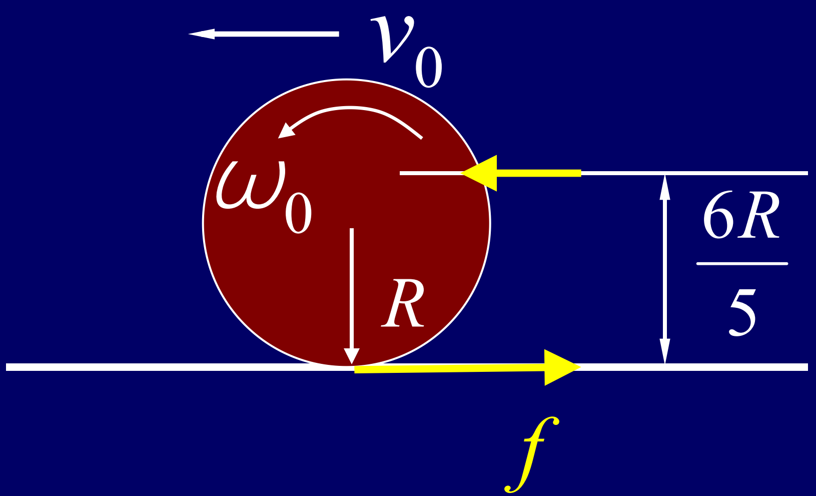
$$\int F dt = mv_0$$

Angular Impulse

$$\int F \frac{R}{5} dt = \frac{2}{5} mR^2 \omega_0$$

$$\int F dt = 2mR\omega_0$$

Example



$$a = -\frac{f}{m} \qquad a = \frac{fR}{\frac{2}{5}mR^2}$$

$$v = v_0 - \frac{f}{m}t \qquad v = R\omega$$

$$v = \frac{v_0}{2} + \frac{f}{\frac{2}{5}m}t$$

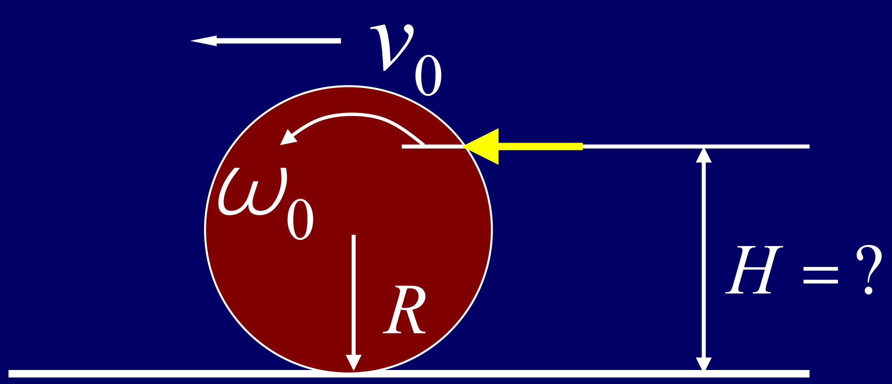
$$\frac{2}{5}v = \frac{v_0}{5} + \frac{f}{m}t$$

$$v = \frac{6}{7}v_0 < v_0$$

$$2mR\omega_0 = mv_0$$

$$2R\omega_0 = v_0$$

Example $v_0 = v = R\omega_0$



$v_0 = R\omega_0$

Impulse

$$\int F dt = m v_0$$

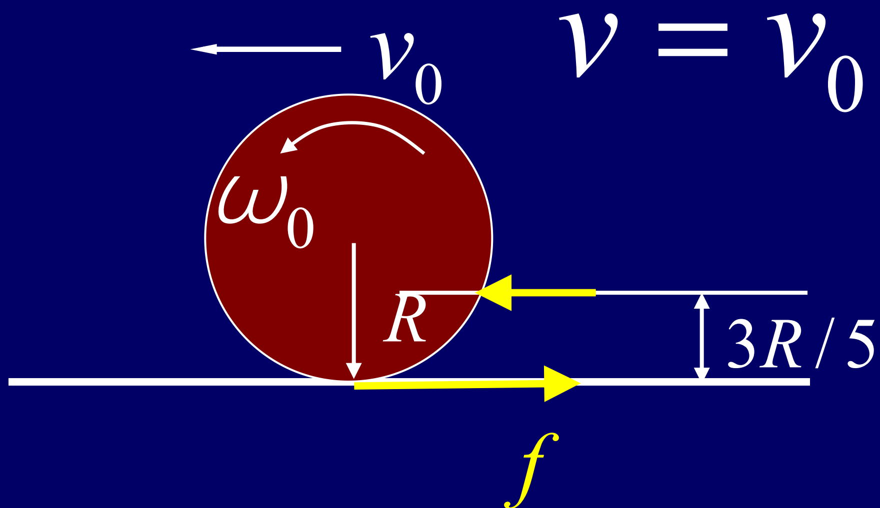
Angular Impulse

$$\int F dt = \frac{2mR^2\omega_0}{5(H-R)}$$

$$m v_0 = \frac{2}{5}\frac{mRv_0}{(H-R)}$$

$$H = \frac{7}{5}R$$

Example



$$v = v_0$$

Impulse

$$\int F dt = m v_0$$

Angular Impulse

$$\int \frac{2RF}{5} dt = \frac{2mR^2}{5} \omega_0$$

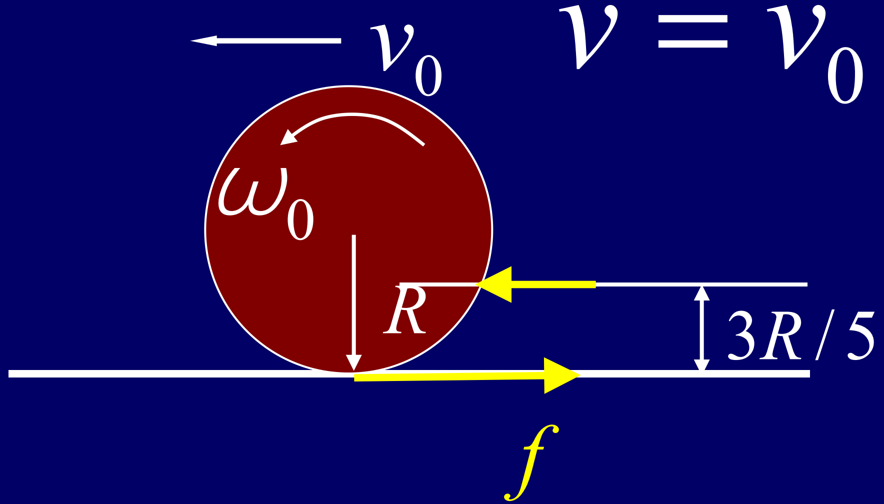
$$\int F dt = mR\omega_0$$

$$R\omega_0 = v_0$$

Example

$$a = -\frac{f}{m} \qquad \alpha = \frac{fR}{\frac{2}{5}mR^2}$$

$$v = v_0$$



$$v = v_0 - \frac{f}{m}t$$

$$\omega = \omega_0 + \frac{fR}{\frac{2}{5}mR^2}t$$

$$\omega R = \omega_0 R + \frac{f}{\frac{2}{5}m}t$$

$$v = \frac{3}{7}v_0 \qquad v = \omega R = -v_0 + \frac{f}{\frac{2}{5}m}t$$

CHAP. 10

Exercises

P224: 1, 3, 4, 9,
15, 20, 21